THE MASTERMIND COACHING CLASSES

Revision Notes

Class 9 · Science

NCERT Curiosity — Chapters 1 to 7 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	Exploration: Entering the World of Secondary Science	Models, language of science, laws and theories, prediction, estimation
2	Exploring Mixtures and their Separation	Solutions, concentration, separation techniques, suspensions, colloids, Tyndall effect
3	The Atomic Foundations of Matter	Laws of chemical combination, Dalton, bonding, formulae, molecular mass
4	Journey Inside the Atom	Atomic models, subatomic particles, atomic and mass number, electronic configuration, valency, isotopes, isobars
5	Describing Motion Around Us	Distance, displacement, speed, velocity, acceleration, graphs, kinematic equations, circular motion
6	How Forces Affect Motion	Force, friction, Newton’s three laws, systems of objects
7	Work, Energy and Simple Machines	Work, work–energy theorem, kinetic and potential energy, conservation, power, pulley, inclined plane, lever

The Mastermind Coaching Classes — Road No. 5, VKIA, Jaipur — 9887258679

CHAPTER 01 · INTRODUCTION

Exploration: Entering the World of Secondary Science

*This opening chapter is not about one topic. It explains **how** secondary science works: scientists simplify nature with models, speak a precise language of symbols and units, build laws and theories from evidence, make testable predictions, and reason with estimates. The two symbols on every page — a **magnifying glass** (careful observation) and a **compass** (direction and purpose) — sum up the message.*

Big Questions

- Why does science use models instead of studying everything in full detail?
- Why must scientific language and units be precise?
- What is the difference between a law, a theory and a principle?
- How do predictions and estimates help us test ideas?

1.1 From Wonder to Deeper Exploration

- In the middle stage, science meant being curious, observing, questioning and finding how things work; ideas of the living and non-living world were linked.
- In the secondary stage the emphasis shifts to **deep exploration**: how observations lead to measurements, how patterns are written as symbols and equations, how **models** are built, and how ideas are tested, revised and sometimes discarded.
- Science is therefore not only *what* we know but *how* we know it.

Symbol on the page numbers	What it stands for
Magnifying glass	Careful observation — noticing patterns and details that are easy to miss
Compass	Direction — choosing suitable models, asking the right questions and knowing the limits of where an idea applies

REMEMBER

Exploration in science is not wandering aimlessly; it is trying to make sense of the world **with care and purpose**.

1.2 Models: Simplifying a Complex World

The natural world is too complex to study in every detail, so science uses **models** — simplified ways of looking at a real system that keep only what matters for a given question. Leaving out details is done **on purpose**; it is not a mistake.

Branch	How a model simplifies
Physics	A moving car is treated as a single point
Chemistry	Atoms and molecules are drawn as spheres, with sticks for bonds
Biology	A cell is shown as a diagram highlighting only the key parts

Branch	How a model simplifies
Earth science	The Earth is treated as a smooth sphere made of distinct layers

- Building a model involves **assumptions** — deliberately ignoring certain details. Example: when studying a falling object we may neglect **air resistance** to see the basic effect of gravity.
- In biology, to understand how the heart pumps blood, individual cells are ignored and the heart is seen as a working system.
- As more detail is added, the model becomes more complex but also more accurate.

▸ Example 1.1 — Modelling a cricket six

Key question: *will the ball cross the boundary without hitting the ground first?*

Keep (important)	Small effect (can ignore in a simple model)	No effect (ignore)
Mass of the ball; speed and direction of the hit	Air resistance; spin; stitching of the seam	Brand of the bat; colour of the ball; grass on the field

▸ Activity 1.1 — Time to cycle home from school

Keep: distance, average speed, stops at traffic lights. Ignore: colour of the cycle, the weather on one day, small speed variations. Ignoring details is useful because it keeps the model simple enough to give an answer.

> **IMPORTANT**
> **Meghnad Saha** (shown on an Indian postage stamp) studied starlight. He did not model every atom or reaction; he treated star matter as a **hot gas** and focused on **temperature, pressure and how atoms form ions**. This simplification explained why the **colour of a star is linked to its temperature**.

1.3 The Language of Science: Symbols, Units and Mathematics

- Everyday words such as **force**, **work** or **reaction** have **specific meanings** in science, because scientific ideas must be communicated clearly and without ambiguity.
- Quantities are written as symbols (mass **m**, velocity **v**, force **F**, current **I**), each with a defined **unit**, so scientists across the world can describe observations, compare results and build ideas together.
- Mathematics is a **language** for expressing relationships between quantities and testing them. An equation is a compact statement of how things are related, not just a calculating tool.
- Good approach: first **understand the situation**, then identify the **relevant quantities**, then use a relation to reason — do not simply memorise equations.
- Examples of use: distance, time and velocity predict where a moving object will be later; mathematical expressions describe rates of chemical reactions, growth of populations and energy changes in a system.

> **IMPORTANT**
> Scientific symbols come from history and **international agreement**, not always from convenience. The speed of light is written **c** from the Latin *celeritas* (speed). It is a physical constant defined to be exactly **299 792 458 m/s**.

▸ Why standard (SI) units matter

- Measurements rest on agreed **international standards**, not local objects or opinions, so a kilogram of rice means the same everywhere; this also gives fairness in trade.

- **Airplane fuel incident:** a flight needed 22 300 kg of fuel, but the crew used the fuel density in **pounds per litre** instead of **kilograms per litre**. The aircraft was about **15 000 litres short**, glided to an emergency landing and was damaged, though nobody was hurt.
- Lesson: use **SI units** consistently to avoid conversions and errors.

1.4 Laws, Theories and Principles

Term	Meaning	Example from the chapter
Law	Describes a regular **pattern** seen in nature, in words or by a mathematical relation	Newton's laws of motion explain the jerk when a bus stops suddenly
Theory	Goes further and **explains why** the pattern occurs, based on evidence gathered over time	Atomic theory explains how molecules are formed
Principle	A broad idea that helps us make sense of a situation	Conservation of energy applied while climbing stairs

IMPORTANT

In science a **theory is not a guess**. It is an explanation based on careful testing and critical examination. Theories remain open to improvement and may change when new evidence appears — this is what makes science reliable.

1.5 Prediction and Testing Ideas

- When laws, theories and models are well established they let us **predict** what will happen under new conditions, sometimes even when an experiment cannot be done.
- Examples: how far a kicked football travels (motion); how much carbon dioxide is produced or how soft baked bread will be (chemical reactions); how breathing changes while running (biology).
- A scientific prediction is a **reasoned expectation based on evidence**, not a guess. If it matches observation, confidence in the idea grows; if it does not, scientists re-examine assumptions, models or measurements.

▸ Example 1.2 — Testing 'It will rain because the clouds look dark'

Good scientific questions ask for **measurable data and past patterns**, not yes/no answers: What was the sky like when it last rained? Was humidity above 80 per cent last time? What is today's wind speed and direction? Is the temperature dropping as before the last rains?

- **Why forecasts go wrong:** weather depends on temperature, pressure, humidity and wind; very tiny differences can grow with time, so forecasts are reliable for hours or a few days but less certain further ahead.
- Even the best theories have limits. Failure of a prediction is not a weakness — it is science's **greatest strength**, because ideas are rejected on **evidence**, never on opinion or belief. No theory is final or beyond question.
- **Viral claim check:** 'food becomes harmful during an eclipse'. An eclipse is only a play of shadows; no physical, chemical or biological mechanism makes food harmful, so the claim fails when tested with simple scientific questions.

1.6 Estimation and Approximate Reasoning

- Understand the situation → identify the quantities that matter → make a **rough estimate** to see whether an answer is reasonable or impossible.
- Estimating builds intuition, helps detect errors and gives confidence. Science values careful reasoning perhaps even more than precise calculation.

▸ Example 1.3 — Litres of air breathed in a day

Step 1 — Breaths per day
About 12–15 breaths per minute at rest; a day has 60 × 24 = 1440 minutes, so about 18 000–22 000, i.e. roughly **20 000 breaths**.

Step 2 — Volume of one breath
About 4–5 breaths fill a party balloon of about 2 litres, so one breath is about **0.5 litre**.

Step 3 — Combine
20 000 × 0.5 ≈ **10 000 litres per day**.

Step 4 — Check
Another route: about 3 balloons a minute × 2 litres × 1440 minutes ≈ 8640 litres, reasonably close to 10 000 litres, so the estimate is sensible.

- **Rice estimate (Ready to Go Beyond):** an adult needs about 2000–2500 kcal a day; find the calories in 100 g of cooked rice and scale up for four people for a month. The answer should be sensible — 100 g is far too little, a few tonnes far too much.
- Approximate answers suit situations like estimating travel time; exact values are needed in situations like medicine doses or fuel loading.

1.7 Branches of Science and Their Connections

- After Grade 10, science divides into **physics, chemistry, biology and earth science**; the 'Next Level Up' boxes in the textbook hint at what each may offer.
- These divisions are **human-made** to organise knowledge. Nature has no such boundaries, and real problems — climate change, medicines, sustainable technology — need ideas from several disciplines plus mathematics, technology, arts and social sciences.
- **Mask example:** physics (particle motion, electrostatic attraction), chemistry (polymer fibres), biology (size and behaviour of viruses) and mathematics (airflow and filtration efficiency) all explain how a mask works.

1.8 Science as a Human Activity

- Science is not just facts, equations and experiments; it is shaped by **curiosity, creativity, collaboration and careful questioning**.
- It grows as people ask questions, test ideas, share results and learn from mistakes, through the work of many people across cultures and generations.
- Scientific thinking helps you understand technology, evaluate information critically and make sense of the world, even if you do not study science after Grade 10.

1.9 Quick Revision Questions

Likely question	Model answer
What is a scientific model? Why are details ignored?	A simplified representation of a real system that keeps only what matters for a question. Details are ignored on purpose so the problem stays simple enough to answer.
Give one example of a model from each of physics, chemistry and biology.	Physics: car as a point. Chemistry: atoms as spheres with bonds. Biology: labelled cell diagram.
Why did Meghnad Saha treat star matter as a hot gas?	To focus on temperature, pressure and ion formation; this explained why a star's colour depends on its temperature.
Why does science use SI units?	They are international standards, so results can be compared and mistakes avoided; the airplane fuel mix-up shows the risk of mixing units.
Why is the speed of light denoted by c?	From the Latin celeritas meaning speed; it is exactly 299 792 458 m/s.
Differentiate a law from a theory.	A law describes a regular pattern; a theory explains why the pattern occurs, on the basis of evidence.
Is a scientific theory just a guess?	No. It is an explanation built on careful testing and critical examination, open to revision with new evidence.
Why are weather forecasts less reliable for the distant future?	Tiny differences in temperature, pressure, humidity or wind grow with time and change the outcome.
How can the claim 'food is harmful during an eclipse' be tested?	Ask what physical change an eclipse causes; since it is only shadow and no mechanism harms food, the claim has no scientific support.
Estimate the litres of air breathed per day.	About 20 000 breaths × 0.5 litre ≈ 10 000 litres.

★ KEY TAKEAWAYS

- Science is about **how** we know, not only what we know — observe with a magnifying glass, move with a compass.
- **Models** simplify reality by deliberately ignoring unimportant details.
- Precise **symbols, SI units and mathematics** form the language of science; mixing units can be dangerous.
- A **law** gives a pattern, a **theory** explains it, a **principle** is a broad guiding idea; theories are evidence-based, not guesses.
- **Predictions** are reasoned expectations; failed predictions drive improvement.
- Use **estimation** to check whether answers are reasonable.
- Branches of science are human divisions; real problems need many branches together.

CHAPTER 02 · CHEMISTRY

Exploring Mixtures and their Separation

*Sugar from sugarcane, malaria tests from a drop of blood, clean water from muddy water — all depend on **separating mixtures**. This chapter classifies mixtures as **solutions, suspensions and colloids**, shows how to express the **concentration** of a solution in percentages, and explains which technique (crystallization, distillation, chromatography, separating funnel, sublimation, centrifugation, coagulation) suits which mixture and why.*

Big Questions

- Why do suspended particles settle in muddy water but not in milk?
- How is evaporation different from boiling?
- Why do we see bright rays of sunlight through gaps in the leaves of a dense tree?

2.1 Classifying Mixtures

- **Homogeneous mixture (solution):** uniform composition throughout — a well-stirred sugar solution is equally sweet in the first and last sip. Examples: vinegar (acetic acid in water), soda water (carbon dioxide in water), air, brass. A solution always remains homogeneous.
- **Heterogeneous mixture:** composition is not uniform; the parts can be seen or settle. Examples: sand in water, oil and water, smoke, fog, iron filings and sulfur.
- Oil and water are **heterogeneous** (they form separate layers).

▸ Activity 2.1 — Salt, chalk powder and milk in water

Step 1 — Prepare
A: a spatula of common salt in 50 mL water. B: a spatula of chalk powder in 50 mL water. C: a few drops of milk in 50 mL water. Stir each well.

Step 2 — Observe
Check whether particles are visible; pass a laser beam through each beaker and look from the side (never look into the beam — it damages the eyes); leave undisturbed; then filter each and look for residue.

Observation	A: Salt + water	B: Chalk + water	C: Milk + water
Particles visible?	No — clear	Yes — cloudy, visible	Not visible, but looks milky
Laser path visible?	No	Yes	Yes
On standing	Nothing settles	Chalk settles at the bottom	Nothing settles
Residue on filter paper	None	Yes	None

IMPORTANT
So the three mixtures are **different types**: A is a **solution**, B is a **suspension**, C is a **colloid**. The differences lie mainly in **particle size**.

2.2 Solutions and Their Concentration

- A solution is formed when a **solute** (substance that dissolves) is mixed with a **solvent** (substance that dissolves the solute). In sugar + water, sugar is the solute and water the solvent.
- **Concentration** = the amount of solute dissolved in a given amount of solvent or solution.
- Right proportion matters: **ORS** (oral rehydration solution) works only when fixed amounts of salt and sugar are used in a fixed amount of water — not every sugary drink is ORS. Too little pesticide fails to protect a crop, too much damages crop, soil and environment.
- ORS was developed and popularised by the Indian paediatrician **Dilip Mahalanabis**; it has saved millions of lives from dehydration caused by diarrhoea and cholera.

▸ Three ways to express concentration (percentage)

Method	Formula	Meaning	Used for
Mass by mass % (% m/m or % w/w)	(Mass of solute ÷ Mass of solution) × 100	Grams of solute in 100 g of solution	Homogeneous and some heterogeneous mixtures; milk powder, spice mixes, nutrition labels
Mass by volume % (% m/v)	(Mass of solute ÷ Volume of solution in mL) × 100	Grams of solute in 100 mL of solution	Medicines, laboratory liquids (5% glucose; 0.9% saline)
Volume by volume % (% v/v)	(Volume of solute ÷ Volume of solution) × 100	mL of solute in 100 mL of solution	Two miscible liquids — perfumes, cosmetics, vinegar

FORMULA

Mass of solution = mass of solute + mass of solvent (not the solvent alone). In industry, weight and mass are used interchangeably, so % m/m and % w/w are numerically equal.

▸ Worked examples

Example	Working	Answer
10 g salt in 90 g water — % m/m?	Mass of solution = 10 + 90 = 100 g; (10 ÷ 100) × 100	10 % m/m
5 g glucose made up to 100 mL — % m/v?	(5 ÷ 100) × 100	5 % m/v
1 mL pesticide in 100 mL spray — % v/v?	(1 ÷ 100) × 100	1 % v/v
Talcum powder with 4 % m/m zinc oxide: zinc oxide in 300 g?	4 ÷ 100 × 300	12 g
2 tablespoons (15 mL each) concentrate in 150 mL juice — % v/v?	Concentrate = 30 mL; (30 ÷ 150) × 100	20 % v/v
Make 5 % v/v vinegar from glacial (100 %) acetic acid	Take 5 mL glacial acetic acid and add water to make 100 mL total (the same ratio for any quantity)	5 mL in 100 mL

Example	Working	Answer
Cake mix: 75 g sugar, 420 g flour, 5 g sodium hydrogencarbonate	Total = 500 g → sugar 15 %, flour 84 %, soda 1 % (m/m)	15 : 84 : 1 %
Brass with 70 % copper in 120 g	Copper = 0.70 × 120; zinc = the rest	84 g Cu, 36 g Zn
20 g sugar in 80 g water (A); 20 g in 100 g (B); 30 g in 80 g (C)	A: 20/100 = 20 %; B: 20/120 = 16.7 %; C: 30/110 = 27.3 %	C is most concentrated (most solute per 100 g solution)

REMEMBER

A saline drip in hospitals is **0.9 % m/v** sodium chloride — 0.9 g of salt in every 100 mL — which is safe for blood and replaces lost body fluids.

2.3 Solubility

- **Solubility** = the maximum amount of solute that dissolves in a fixed quantity of solvent (100 g or 100 mL) at a given temperature. A solution that cannot dissolve any more solute at that temperature is a **saturated solution**.
- The temperature is always mentioned because solubility depends on it. For **solids in liquids**, solubility generally **increases** with temperature; for **gases in liquids**, it generally **decreases** with temperature (which is why a warm soda goes flat).
- A graph of solubility (g per 100 g water, y-axis) against temperature (°C, x-axis) is a **solubility curve**. A steeper curve means solubility changes more with temperature.

▸ Reading solubility data (Table 5.4)

Salt (g per 100 g water)	10 °C	20 °C	30 °C	40 °C	60 °C	80 °C	Trend
Potassium nitrate	21	32	45	62	106	167	Rises very steeply
Sodium chloride	36	36	36.3	36.5	37	37	Almost unchanged
Potassium chloride	35	35	37.4	40	46	54	Rises moderately
Ammonium chloride	24	37	41	41	55	66	Rises steadily

- Saturated KNO_3 solution in 50 g water at 40 °C needs 62 ÷ 2 = **31 g**.
- Cooling a saturated KCl solution from 80 °C to 25 °C: solubility falls from 54 to about 35 g, so about 19 g per 100 g water separates out as **crystals**.
- Because NaCl's solubility hardly changes with temperature, cooling cannot crystallize it effectively — salt is obtained by **evaporation** instead.

2.4 Separating Homogeneous Mixtures

▸ A. Crystallization

- A **crystal** is a solid whose particles are arranged in a regular geometric pattern (rock salt, sugar crystals/mishri, snowflakes, frost, quartz, copper sulfate).
- **Crystallization** = forming crystals from a hot saturated solution on cooling. **Principle:** solubility of a substance differs at different temperatures. Example: a saturated solution of compound B made at 60 °C

(287 g per 100 g water) when cooled to 40 °C can hold only 241 g, so the extra ~46 g separates as pure solid.

- Used to separate two solids soluble in the same solvent when one is present in small amount, and to **purify** solids (separating a desired pure substance from unwanted impurities).

Step 1 — Saturate
Dissolve about 1 g copper sulfate in 25 mL water in a beaker with a drop of dilute sulfuric acid (added by the teacher; it helps give pure crystals); warm in a water bath, stirring, adding more copper sulfate until no more dissolves.

Step 2 — Filter hot
Filter the hot saturated solution to remove insoluble impurities; collect the filtrate in a clean beaker and cover with a watch glass.

Step 3 — Cool slowly
Leave undisturbed. Slow cooling gives particles time to join, forming **larger, shiny, well-shaped blue crystals**; rapid cooling (ice-cold water) gives small, poorly formed crystals.

Step 4 — Collect
Filter the crystals, rinse with cold water and dry on a watch glass. Copper sulfate is toxic — work under adult supervision.

- **Salt from seawater:** seawater is held in shallow ponds, the sun evaporates water, a saturated solution forms and salt crystals deposit. Ancient Indian methods: *panga* salt (boiling concentrated brine) and *karkatch* salt (solar evaporation of sea water) — giving crystals of different sizes.
- Natural crystals: Mawsmai Cave (Sohra/Cherrapunji) formations and quartz.

	Evaporation	Crystallization	Distillation
What is obtained	Solute only (solvent lost)	Pure crystals of the solute	Solvent/liquid (also a liquid from a liquid mix)
Process	Solvent vaporises, solid left behind	Hot saturated solution is cooled; crystals form	Heat, vaporise lower-boiling liquid, condense
Prefer when	Only the solid is needed; it is not affected by heat (e.g. salt)	Pure, well-formed solid needed; solubility changes with temperature; solid may decompose on heating	The solvent or the lower-boiling liquid is also to be recovered

▸ B. Distillation

- A homogeneous mixture of **two miscible liquids** is heated until the liquid with the **lower boiling point** vaporises; its vapour is cooled in a **condenser** and collected as the pure **distillate**. The other liquid (or dissolved solid) stays in the distillation flask.
- Works when boiling points differ by **at least about 25 °C**. Also used to recover a liquid from a solution containing dissolved solids (e.g. pure water from salt water).
- **Set-up:** distillation flask with thermometer on a tripod/wire gauze and burner → water condenser (water enters at the bottom inlet, leaves from the top outlet) → conical flask to collect distillate.
- Acetone (b.p. 56 °C) and water (100 °C) separate easily: acetone distils first.

Compound	Water	Acetone	Alcohol	Chloroform	Benzene
Boiling point	100 °C	56 °C	78 °C	61 °C	80 °C

- With Table 5.5: water–acetone (44 °C gap) can be separated by simple distillation; acetone–alcohol (22 °C), alcohol–chloroform (17 °C) and alcohol–benzene (2 °C) have gaps smaller than 25 °C and need

fractional distillation; sand–salt are two solids and need dissolving/filtering/evaporation.

- **Fractional distillation** separates liquids with small boiling-point differences (less than 25 °C). Crude petroleum is separated in a refinery into petroleum gas (filled in cylinders and liquefied as **LPG**), petrol, aviation fuel (kerosene), diesel, lubricating oil and bitumen.
- **Kannauj (Uttar Pradesh)**, the perfume capital of India, captures the smell of first rain as *mitti ka ittar* using the traditional **Deg-Bhapka** distillation method.

▸ C. Paper chromatography

- The word comes from Greek *chroma* (colour) and *graphein* (to write) — first used to separate dyes and inks.
- **Principle:** components differ in how strongly they interact with the paper and the solvent, so they travel at **different rates** and separate into spots.

Step 1 — Spot
On a 3 cm wide chromatography (or filter) paper strip draw a pencil line 2 cm from the bottom; put a spot of black sketch-pen ink at its centre.

Step 2 — Develop
Dip the lower end in a thin layer of water in a gas jar/beaker, hanging vertically. The water level must be **below the spot**.

Step 3 — Observe
As water rises it carries the colours along; the ink splits into different coloured spots. Faster-moving colours travel higher.

- Can be applied to green food colour (with 2 % m/v salt solution as solvent), spinach pigments, flower pigments, fennel-seed coloured freshener. Water does not suit every case — alcohol or a solvent mixture may be needed.
- A pencil (not ink) is used for the baseline because graphite does not dissolve or move with the solvent.

2.5 Separating Heterogeneous Mixtures

▸ A. Two immiscible liquids — separating funnel

- Immiscible liquids (oil and water) form two layers and are separated by **density difference**.
- Pour the mixture into a separating funnel and let it stand. Mustard oil (less dense) forms the **upper** layer; water the **lower**. Open the stopcock slowly to drain the water; close it when nearly all water is out; collect the in-between mixed portion separately and discard it; then collect the oil.
- Cooking oil (910 g per litre) is lighter than water (1000 g per litre), so it floats on top. If two immiscible liquids had the **same density**, they would not form distinct layers and the funnel would fail.
- Mixtures with gases: gases mix freely, so most gas mixtures are homogeneous (air, hydrogen + oxygen rocket fuel). **Smoke** (solid in air), **fog** (liquid droplets in air) and dust in air are heterogeneous.

▸ B. Sublimation (solid–solid mixtures)

- **Sublimation:** a solid changes directly to vapour on heating, below its melting point, without passing through the liquid state. **Deposition:** vapour cools and turns directly into solid.
- **Activity:** heat a camphor–sand mixture gently in a china dish with an inverted funnel (nozzle plugged with cotton) on top. Camphor sublimes and deposits as white solid on the cool funnel wall; sand stays behind. Naphthalene and ammonium chloride behave similarly; dry ice (solid CO_2) also sublimes.
- Sublimation differs from evaporation: evaporation is liquid → vapour; sublimation is solid → vapour directly.

- Sequence for sand + salt + naphthalene: (1) **sublimation** removes naphthalene; (2) add water and **filter** to leave sand; (3) **evaporate/crystallize** the filtrate to get salt.

▸ Alloys

- Metals do not dissolve in each other at room temperature, but molten metals can mix into a solution which solidifies on cooling. An **alloy** is a homogeneous mixture of two or more metals, or of a metal and a non-metal. Its components **cannot be separated by physical methods**.
- Made to get materials that are stronger, more rigid or more corrosion-resistant.

Alloy	Composition
Brass	About 80 % copper, 20 % zinc
Bronze	About 80 % copper, 20 % tin
Stainless steel	Iron with carbon (0.03–0.8 %), chromium (16–18 %), nickel (10–14 %), molybdenum (2–3 %)

2.6 Suspensions

- A **suspension** is a heterogeneous mixture in which insoluble solid particles remain spread (suspended) through the medium. Particles are **visible to the naked eye** and larger than 1000 nm; they settle on standing. Examples: sand, sawdust or tea leaves in water; muddy water.
- Filtering through cloth or filter paper removes only larger particles — the water often stays cloudy, so further methods are needed.

▸ Centrifugation

- Spinning a mixture in a tube at high speed: the tubes become horizontal and **centrifugal force** (the outward force on a body moving in a circle) pushes heavier particles to the bottom; the lighter liquid remains on top. (Feel it in the partner-spinning game *phugadi*, a folk dance.)
- Used in labs to separate blood into **red blood cells, white cells, platelets and plasma**, and in chemical industries.
- **Paperfuge:** a hand-powered device based on a toy (cardboard disc, string, handles) that separates blood without electricity; helps detect malaria and anaemia in remote areas.
- Blood donation: donated blood is tested, its group identified and it is separated into plasma, platelets, white and red cells for storage in blood banks; the body replaces donated blood in a few weeks.

▸ Coagulation

- Adding **alum (fitkari)** to muddy water makes fine suspended particles clump together; the alum is the **coagulant**. The larger clumps settle by gravity (**sedimentation**) and the clear water is then separated by **decantation** or filtration.
- Everyday example: making **paneer** — lemon juice or vinegar (acid) coagulates milk proteins.
- Sewage treatment uses sedimentation, coagulation and filtration; the clean water can be reused for flushing or watering plants.

2.7 Colloids and the Tyndall Effect

- Blood, milk, tomato sauce and ice cream are **colloids** — neither true solutions nor suspensions. Particle size is **1–1000 nm**; the particles are too small to see with the naked eye, do **not settle**, and are uniformly dispersed (so a colloid looks homogeneous).
- Components: the **dispersed phase** (the solute-like particles) and the **dispersion medium** (the substance in which they are spread). In blood sampled for centrifugation the cells/platelets are the

dispersed phase and plasma is the medium. If blood behaved like a true suspension, the cells would settle and could not circulate to carry nutrients, gases and hormones.

- **Emulsions:** colloids in which both phases are liquids. *Oil-in-water* (milk, vanishing cream); *water-in-oil* (butter, cold cream, body lotion). **Emulsifying agents** (e.g. milk proteins) stabilise them. Shake oil with soap solution to make one.
- Clouds (tiny water droplets/ice crystals in air) are colloids; smoke and dust make city air hazy by scattering light.

DEFINITION

Tyndall effect: the scattering of light by particles in a medium, making the path of a light beam visible. Named after John Tyndall. Seen in a sunbeam entering a dark room (dust/smoke), through forest leaves, and in stadium floodlights.

Property	Solution	Suspension	Colloid
Nature	Homogeneous	Heterogeneous	Appears homogeneous, actually heterogeneous
Particle size	Less than 1 nm	More than 1000 nm	1–1000 nm
Visibility	Particles not visible; transparent	Particles visible; opaque/cloudy	Particles not visible; translucent/milky
Filtration	Passes through filter paper	Residue stays on filter paper	Passes through filter paper
Settling on standing	No	Yes	No
Tyndall effect	No	Yes	Yes
Examples	Salt solution, brass, vinegar, air	Sand/mud in water, chalk in water	Milk, butter, smoke, blood, tomato sauce

IMPORTANT

Assertion–reason favourite: *Solutions do not show the Tyndall effect* is true, but the reason is **not** that solute particles are larger than 100 nm — they are **smaller than 1 nm** and too small to scatter light. Mixtures showing the effect: air with dust and starch in water; not copper sulfate or acetone in water.

2.8 Separation in Real Life

- Kidneys separate waste from blood; sewage treatment, ocean/river plastic clean-up and **waste segregation** (dry waste recycled, wet waste composted) all rely on separation methods.
- Researchers recover valuable materials such as **lithium** from old mobile-phone and laptop batteries.

Mixture	Method	Reason
Mud from muddy water	Sedimentation, decantation/filtration (coagulation if fine)	Insoluble particles larger than water
Plasma from blood	Centrifugation	Different densities
Naphthalene and sand	Sublimation	Naphthalene sublimes, sand does not

Mixture	Method	Reason
Chalk powder and salt	Add water, filter, evaporate filtrate	Salt dissolves, chalk does not
Salt and water	Evaporation (or distillation to also get water)	Salt is non-volatile
Oil and water	Separating funnel	Immiscible; different densities
Pigments of a flower	Paper chromatography	Different rates of movement on paper
Acetone (60 °C) and another liquid (90 °C) — miscible	Simple distillation	Boiling points differ by 30 °C (>25 °C)

Likely question	Model answer
Differentiate between a solution, suspension and colloid.	Particle size: <1 nm, >1000 nm, 1–1000 nm. Solutions are clear, do not settle, no Tyndall effect; suspensions settle, are filtered and scatter light; colloids do not settle, pass filter paper and scatter light.
Define Tyndall effect and give two examples.	Scattering of light by particles in a medium so that the beam's path is visible; sunbeam in a dusty room, headlight beam in fog or stadium floodlights.
Write the three formulae for concentration in percentage.	% m/m = (solute mass ÷ solution mass) × 100; % m/v = (solute mass ÷ solution volume) × 100; % v/v = (solute volume ÷ solution volume) × 100.
What is solubility? How does temperature affect it?	Maximum solute dissolving in 100 g solvent at a given temperature; solids' solubility generally increases, gases' decreases with temperature.
Why is slow cooling preferred in crystallization?	It lets particles arrange gradually, giving large, well-shaped, pure crystals.
When is distillation used? What is the condition?	To separate miscible liquids (or a liquid from a dissolved solid); boiling points must differ by at least ~25 °C; otherwise use fractional distillation.
Explain paper chromatography.	Solvent rises through paper, carrying components at different rates because of different interactions with paper and solvent, forming separate spots; the baseline spot must stay above the solvent.
Why does oil float on water and how are they separated?	Oil is less dense; use a separating funnel and drain the lower water layer first.
What is sublimation? Name two substances that sublime.	Solid to vapour directly on heating; camphor, naphthalene (also ammonium chloride, dry ice).
How do centrifugation and coagulation differ?	Centrifugation uses rapid spinning (centrifugal force) to settle heavier particles; coagulation uses a coagulant such as alum to clump fine particles which then settle.

Likely question	Model answer
What is an alloy? Give composition of brass.	Homogeneous mixture of metals (or a metal and non-metal); brass ≈ 80 % copper, 20 % zinc.
What is an emulsion? Give one example of each type.	Colloid of two liquids; milk (oil-in-water), butter (water-in-oil).

★ KEY TAKEAWAYS

- Mixtures are homogeneous (solutions) or heterogeneous; by particle size they are solutions (<1 nm), colloids (1–1000 nm) or suspensions (>1000 nm).
- Concentration is expressed as % m/m, % m/v or % v/v; mass of solution = solute + solvent.
- Solubility of solids rises with temperature, of gases falls; crystallization exploits this.
- Distillation (≥25 °C difference), fractional distillation, chromatography, separating funnel, sublimation, centrifugation and coagulation each suit a specific kind of mixture.
- Colloids and suspensions scatter light (Tyndall effect); true solutions do not.
- Alloys are homogeneous mixtures that cannot be separated physically.

CHAPTER 03 · CHEMISTRY

The Atomic Foundations of Matter

*Why does a compound behave so differently from the elements that formed it, yet weigh exactly as much as they did together? This chapter builds the foundations of chemistry: the **Law of Conservation of Mass**, the **Law of Constant Proportions**, **Dalton's atomic theory**, how atoms bond by **sharing (covalent)** or **transferring (ionic)** electrons, how to **name and write formulae**, and how to calculate **molecular mass** and **formula unit mass**.*

Big Questions

- Is water from every source chemically identical?
- Why is oxygen written O in some places and O_2 in others?
- Why does salt solution conduct electricity but sugar solution does not?

3.1 Mass in Physical and Chemical Changes

- Hydrogen (combustible) and oxygen (supports burning) combine to form water, which does neither — the properties of a compound differ from its elements — yet the **mass of water formed equals the sum of the masses of hydrogen and oxygen**.
- **Activity 3.1 (physical change):** weigh water + salt on a digital balance (tare set to zero first), then swirl until dissolved. The mass is unchanged — no change in mass in a physical change (also tearing paper). Variation of ±1 in the last digit is experimental error.

▸ Activity 3.2 — Baking soda and vinegar

	Set-up 1 (open)	Set-up 2 (closed with balloon)
Arrangement	Vinegar in flask; baking soda in balloon placed beside it; baking soda then poured in	Balloon with baking soda fixed on mouth of flask holding vinegar; soda tipped in
Observation	Brisk effervescence; final reading **less** than initial	Balloon inflates; final reading **equals** initial
Reason	CO_2 gas escapes into the air	CO_2 is trapped in the balloon, nothing escapes

REMEMBER

Vinegar + baking soda (sodium hydrogencarbonate) → carbon dioxide + other substances. Mass appears to drop only when a gaseous product escapes. Tip: keep the flask and balloon on the balance throughout so stray traces do not cause errors.

3.2 Law of Conservation of Mass

DEFINITION

Law of Conservation of Mass (Lavoisier, 1789): Matter can neither be created nor destroyed in a chemical reaction — the total mass of reactants equals the total mass of products. Lavoisier is called the **Father of Modern Chemistry**.

- **Activity 3.3:** 10 mL of 1 % m/v sodium sulfate (flask A) and 10 mL of 1 % m/v barium chloride (flask B) are weighed together, then mixed. A white precipitate of **barium sulfate** forms along with sodium chloride: sodium sulfate + barium chloride → barium sulfate + sodium chloride. The total mass is unchanged. Keep both flasks on the balance to avoid loss from solution sticking to walls.
- Zinc + dilute HCl → zinc chloride + hydrogen: in an open container hydrogen escapes; to test the law you must use a closed system.
- Burning 10 g ethanol in an open beaker leaves no residue, but the law is **not violated**: the products (CO_2 and water vapour) escape into the air, and total mass of reactants (ethanol + oxygen) equals total mass of products.

▸ Numericals

Problem	Working	Result
4.0 g $CaCO_3$ + 2.92 g HCl → 1.76 g CO_2 + 0.72 g H_2O + 4.44 g $CaCl_2$	Reactants 4.0 + 2.92 = 6.92 g; products 1.76 + 0.72 + 4.44 = 6.92 g	Law obeyed
12 g C + 32 g O_2 → 44 g CO_2; how much CO_2 from 2.4 g C?	44/12 × 2.4	8.8 g
20 g H_2 + 160 g O_2 → water	20 + 160	180 g water
5.3 g Na_2CO_3 + 6.0 g acetic acid → 2.2 g CO_2 + 0.9 g H_2O + 8.2 g sodium acetate	Left 11.3 g; right 2.2 + 0.9 + 8.2 = 11.3 g	Law valid

3.3 Law of Constant (Definite) Proportions

DEFINITION

Law of Constant Proportions (Proust's law): In any chemical compound, the elements are always present in a **fixed ratio by mass**, irrespective of source or method of preparation.

- Water from rivers, borewells or the ocean, when purified and decomposed, always gives hydrogen and oxygen in the mass ratio **1 : 8** (9 g water → 1 g H + 8 g O).
- Proust showed copper carbonate always has Cu : C : O in fixed proportion. Cinnabar (*hingula*), the red ore known in many ancient civilisations, yields mercury (≈ 86.22 %) and sulfur (≈ 13.78 %) on heating; grinding them in this ratio re-forms it (though toxicity limited this).

Problem	Working	Answer
NaCl has Na : Cl = 23 : 35.5. Chlorine needed for 46 g Na?	35.5 ÷ 23 × 46	71 g
Compound 40 % S, 60 % O. Oxygen with 20 g S?	S : O = 2 : 3, so 20 × 3/2	30 g
CO has C : O = 3 : 4. Oxygen for 9 g C?	9 × 4/3	12 g
Copper oxide made as 4 : 1 (X) and 8 : 2 (Y)	8 : 2 simplifies to 4 : 1	Same ratio — law is supported

IMPORTANT

The law holds for **compounds** but not for **mixtures**, whose components can be mixed in any proportion (e.g. sugar and water in a solution).

3.4 Dalton's Atomic Theory

- The two laws were explained by **John Dalton** (taught in Manchester; presented his theory in **1808**). A **postulate** is a fundamental assumption accepted without proof from which ideas are developed.

1. All matter is made up of very tiny particles called **atoms**, which take part in chemical reactions.
2. Atoms are **indivisible** and cannot be created or destroyed in a chemical reaction.
3. Atoms of a given element are **identical** in mass and chemical properties.
4. Atoms of different elements have **different** masses and chemical properties.
5. Atoms combine in the ratio of **simple whole numbers** to form compounds.
6. The relative number and kinds of atoms are **constant** in a given compound.

Postulate	Explains
Atoms are neither created nor destroyed, only rearranged	Law of Conservation of Mass
Fixed number and kind of atoms in a compound	Law of Constant Proportions

- Examples: magnesium burning in air gives white magnesium oxide (Mg atoms combine with O atoms). H and O atoms combine to form water without being destroyed.
- Later work showed atoms are divisible (electrons, protons, neutrons) — see Chapter 4. Assertion–reason: 2 g H + 16 g O → 18 g water is true (ratio 1 : 8) and Dalton's whole-number combining of atoms explains it, so both true and R is the correct explanation.

3.5 How Atoms Combine

- A **molecule** is an electrically neutral entity of more than one atom that can exist independently and shows all the properties of that substance. Molecules of elements (H_2, O_2) have atoms of one kind; molecules of compounds (HCl, H_2O) have different kinds. Noble gases such as helium exist as single atoms because they are already stable.
- Stability = full valence shell: **octet** (8 electrons) or **duplet** (2 when K is the outer shell). Atoms with fewer electrons **share, gain or lose** electrons — in two ways: **sharing** (covalent) and **transfer** (ionic).
- Combination lowers the energy of the system below that of separate atoms, which makes it stable. The force holding atoms together is a **chemical bond**.

▸ Covalent bond — bonding by sharing

Molecule	Valence electrons needed	Sharing	Bond	Structure
Hydrogen H_2	1 each (to complete duplet)	1 pair	Single	H—H
Chlorine Cl_2	1 each (has 7)	1 pair	Single	Cl—Cl
Oxygen O_2	2 each (has 6)	2 pairs	Double	O=O
Nitrogen N_2	3 each (has 5)	3 pairs	Triple	N≡N
Hydrogen chloride HCl	H needs 1, Cl needs 1	1 pair	Single	H—Cl

Molecule	Valence electrons needed	Sharing	Bond	Structure
Water H_2O	O needs 2, each H needs 1	O shares with two H atoms	2 single	H—O—H

- A **covalent bond** is formed by a shared pair of electrons that attracts both nuclei. Other examples: CO_2 (O=C=O), NH_3 (N bonded to three H), H_2S, F_2 (fluorine, atomic number 9, 2,7: one shared pair).
- Neon (2, 8) neither shares nor transfers electrons because its octet is already complete.

▸ Naming covalent compounds

- Use prefixes for the number of atoms: mono-1, di-2, tri-3, tetra-4, penta-5, hexa-6. The first element keeps its name (no *mono-* on it); the second ends in **-ide**.
- Drop the vowel of the prefix before an element starting with a vowel (**monoxide**, **pentoxide**) but keep 'i' (**dioxide**, **trioxide**). Hydrogen as the first element never takes a prefix (H_2S = hydrogen sulfide).
- Some compounds keep common names: H_2O (water), NH_3 (ammonia).

Formula	Name	Formula	Name
CO	Carbon monoxide	SF_6	Sulfur hexafluoride
CO_2	Carbon dioxide	N_2O_4	Dinitrogen tetroxide
CS_2	Carbon disulfide	N_2O_5	Dinitrogen pentoxide
PCl_3	Phosphorus trichloride	NO_2	Nitrogen dioxide

▸ Ionic bond — bonding by transfer

- Atoms with **fewer than 4** valence electrons usually **lose** them; those with **more than 4** usually **gain or share**.
- **NaCl:** sodium (2, 8, 1) loses 1 electron → **Na^+ cation** (11 protons, 10 electrons). Chlorine (2, 8, 7) gains 1 → **Cl^- anion**. Oppositely charged ions attract by an electrostatic force — the **ionic bond**.
- Sulfur gains 2 electrons → S^{2-}; oxygen → O^{2-}; magnesium loses 2 → Mg^{2+}. Mg^{2+} + 2 Cl^- → $MgCl_2$; 2 Na^+ + S^{2-} → Na_2S; Ca → Ca^{2+}, K → K^+.
- Ionic compounds form **3-D crystals** (crystal lattice): in NaCl each Na^+ is surrounded by six Cl^- and each Cl^- by six Na^+. They do not form separate molecules.
- Ionic compounds generally form between **metals and non-metals**. Name the cation first, then the anion (simple anions end in -ide, polyatomic ions usually do not).

Covalent bond	Ionic bond
Sharing of electrons	Transfer of electrons
Between non-metal atoms	Between a metal and a non-metal
Neutral molecules (H_2O, HCl)	Ions in a 3-D crystal (NaCl)

> **IMPORTANT**
> Nuclear (atomic) energy arises when nuclei split or combine — used in power generation, medicine, research and space. **Raja Ramanna** is called the Father of the Indian Nuclear Programme (peaceful use).

3.6 Valency and Writing Chemical Formulae

Cations	Valency	Anions	Valency
Na^+, Li^+, K^+, Ag^+, Cu^+ (cuprous), NH_4^+	1	F^-, Cl^-, Br^-, I^-, OH^-, NO_3^-, HCO_3^-	1
Ca^{2+}, Ba^{2+}, Mg^{2+}, Zn^{2+}, Fe^{2+} (ferrous), Cu^{2+} (cupric)	2	O^{2-}, S^{2-}, CO_3^{2-}, SO_4^{2-}	2
Al^{3+}, Fe^{3+} (ferric)	3	PO_4^{3-} (used in exercises)	3

Step 1
Write the symbol of the cation (or the first element) and then the anion.

Step 2
Write their valencies/charges below the symbols (numbers only).

Step 3
Criss-cross the numbers as subscripts.

Step 4
Reduce to the simplest ratio by a common factor; omit subscript 1; use **brackets** when two or more identical polyatomic ions are present.

Compound	Working	Formula
Hydrogen sulfide	H 1, S 2	H_2S
Carbon tetrachloride	C 4, Cl 1	CCl_4
Calcium chloride	Ca 2, Cl 1	$CaCl_2$
Aluminium oxide	Al 3, O 2	Al_2O_3
Magnesium oxide	Mg_2O_2 reduced	MgO
Calcium carbonate	$Ca_2(CO_3)_2$ reduced	$CaCO_3$
Magnesium hydroxide	Mg 2, OH 1	$Mg(OH)_2$
Aluminium hydroxide	Al 3, OH 1	$Al(OH)_3$ (not $AlOH_3$)
Aluminium sulfate	Al 3, SO_4 2	$Al_2(SO_4)_3$
Sodium hydrogencarbonate	Na 1, HCO_3 1	$NaHCO_3$
Ferric chloride	Fe 3, Cl 1	$FeCl_3$
Cuprous oxide	Cu 1, O 2	Cu_2O
Aluminium nitrate	Al 3, NO_3 1	$Al(NO_3)_3$
Potassium carbonate	K 1, CO_3 2	K_2CO_3
Ferric hydroxide	Fe 3, OH 1	$Fe(OH)_3$
Magnesium nitride	Mg 2, N 3	Mg_3N_2

3.7 Properties of Ionic and Covalent Compounds

Property	Ionic compounds (NaCl, $CuSO_4$)	Covalent compounds (camphor, naphthalene, sugar)
Solubility	Generally soluble in water; insoluble in kerosene and petrol	Most insoluble in water (sugar is an exception); dissolve in kerosene, petrol
Conductivity (solid)	Do not conduct — ions fixed in the lattice	Do not conduct
Conductivity (water solution)	Conduct — free ions	Do not conduct, even sugar, since no ions form
Molten state	Conduct — ions become free	Do not conduct
Melting and boiling points	High — strong inter-ionic attraction	Usually low

- A solid that does not conduct but conducts in aqueous solution has an **ionic bond**.
- Metal M with two valence electrons forms MO (e.g. Mg: MgO), ionic, whose aqueous solution conducts electricity.
- Assertion–reason: copper sulfate conducts in molten but not solid state — A is true but R is false (ions are *fixed* in solid, *free* when molten).

3.8 Molecular Mass and Formula Unit Mass

FORMULA

Molecular mass (covalent compounds) = sum of atomic masses of all atoms in the molecule, in **u**.
Formula unit mass (ionic compounds) = sum of atomic masses in one formula unit (simplest ratio of ions; ionic compounds form no molecules).

Compound	Calculation	Mass
H_2O	(1 × 2) + 16	18 u
CO_2	12 + (16 × 2)	44 u
HNO_3	1 + 14 + 48	63 u
CH_4	12 + 4	16 u
Na_2O	(23 × 2) + 16	62 u
$Ca(NO_3)_2$	40 + 2 × (14 + 48)	164 u
KCl	39 + 35.5	74.5 u
$Mg(OH)_2$	24 + 2 × (16 + 1)	58 u
NH_4NO_3	14 + 4 + 14 + 48	80 u
H_3PO_4	3 + 31 + 64	98 u
$NaHCO_3$	23 + 1 + 12 + 48	84 u

- Particles in atoms: Al-27 has 13 electrons and 14 neutrons; Br-80 has 35 electrons and 45 neutrons; Hg^{2+} (mass 201) has 78 electrons and 121 neutrons.
- A species with 11 protons, 12 neutrons and 10 electrons: atomic number 11, mass number 23, a **cation** (Na^{+}), configuration 2, 8.

Likely question	Model answer
State the Law of Conservation of Mass. Who proposed it?	Mass is neither created nor destroyed in a chemical reaction; Antoine Lavoisier (1789).
Why did the mass fall in the open baking soda–vinegar experiment?	The CO_2 gas escaped; in the closed balloon set-up the mass stayed constant.
State the Law of Constant Proportions with an example.	A compound always has its elements in a fixed mass ratio; water always has H : O = 1 : 8.
List Dalton's postulates.	Atoms are tiny, indivisible, not created or destroyed; identical for an element, different for different elements; combine in simple whole-number ratios; fixed number and kind in a compound.
Distinguish covalent and ionic bonds.	Covalent: sharing of electron pairs between non-metals; ionic: transfer of electrons from a metal to a non-metal giving ions held by electrostatic attraction.
Why do ionic compounds conduct only in solution or molten state?	Ions are fixed in the solid lattice and become free to move only on dissolving or melting.
Why does O_2 have a double bond and N_2 a triple bond?	O needs 2 electrons, N needs 3 to complete the octet, so 2 and 3 pairs are shared.
Write the formula of aluminium sulfate and explain.	Al^{3+} and SO_4^{2-} criss-cross to $Al_2(SO_4)_3$; brackets for 3 sulfate ions.
Name N_2O_5 and SF_6.	Dinitrogen pentoxide; sulfur hexafluoride.
Calculate the molecular mass of CO_2 and the formula unit mass of $Ca(NO_3)_2$.	44 u; 164 u.
What is a formula unit?	The simplest whole-number ratio of ions in an ionic compound (e.g. NaCl, Na_2O).
How are cations and anions formed from Na and Cl?	Na loses one valence electron to become Na^{+}; Cl gains one to become Cl^{-}.

★ KEY TAKEAWAYS

- Mass is conserved in every chemical reaction (Lavoisier); watch for escaping gases in open systems.
- Compounds have elements in a fixed mass ratio (Proust) — water is always H : O = 1 : 8.
- Dalton’s theory explains both laws; atoms are rearranged, not created or destroyed.
- Covalent bonds share electron pairs (single, double, triple); ionic bonds transfer electrons and form 3-D crystals.
- Write formulae by criss-crossing valencies, simplifying and using brackets for repeated polyatomic ions.
- Ionic compounds: high melting point, conduct in solution/molten state; covalent: low melting point, usually non-conducting.
- Molecular mass (covalent) and formula unit mass (ionic) are sums of atomic masses.

CHAPTER 04 · CHEMISTRY

Journey Inside the Atom

*Everything — living beings and a house alike — is made of atoms, yet the atom itself is not the end of the story. This chapter follows the **history of atomic models** (Kanada and Democritus → Dalton → Thomson → Rutherford → Bohr), the discovery of the **electron, proton and neutron**, and then the language used to describe atoms: **symbols, atomic number, mass number, electronic configuration, valency, isotopes and isobars**.*

Big Questions

- Are atoms the smallest indivisible particles?
- Why do electrons not fall into the nucleus although protons attract them?
- Why did scientists keep modifying atomic models?

4.1 Rediscovering the Roots of Atomic Theory

- More than 2,000 years ago, thinkers in India and Greece asked: *what is everything made of?*
- **Acharya Kanada** (Vaisesika Sutras): dividing matter (*dravya*) repeatedly gives the smallest indivisible particles, **parmanus**, which cannot be sensed; they combine into **dyads** (two) and **triads** (three) to make the whole universe. The proportions of combination were not specified.
- **Leucippus and Democritus** (Greece): indivisible particles called **atomos** (Greek: *indivisible*).
- These ideas were imaginary and philosophical, not based on experiments. In **1808 John Dalton** proposed the first scientific atomic theory based on experiments: matter is made of indivisible atoms — the starting point for atomic structure.
- Later questions: what are atoms made of? what do they look like? what makes atoms of one element differ from another?

4.2 Historical Journey of Atomic Models

- Late 19th century: **radioactivity** (some elements emit radiation) showed that atoms contain smaller particles, so atoms are *not* indivisible.
- Colours in atom diagrams are only illustrative; atoms have no colour.

▸ Discovery of the electron (J. J. Thomson, 1897)

- He passed high voltage through gas at very low pressure in a glass tube with two electrodes. Rays moved from the **cathode (−)** to the **anode (+)** — **cathode rays**.
- In electric and magnetic fields they behaved as streams of **negatively charged particles**, much lighter than atoms — **electrons**.
- Cathode rays were the same whatever the cathode material or gas, so electrons are present in the atoms of **every** element.
- Charge on an electron = -1.602×10^{-19} C, taken as **−1** by convention. Thomson won the 1906 Nobel Prize in Physics and headed the Cavendish Laboratory, Cambridge.

▸ Thomson's model (plum pudding / watermelon model)

- Atom = a sphere of **positive charge** with electrons embedded throughout (plums in pudding; seeds in a watermelon). It explained how an atom stays neutral.

- An orange with seeds is a similar analogy; it falls short because an atom's positive charge is not a soft pulp and the electrons carry equal and opposite charge in a balanced amount.

▸ Gold foil experiment (Geiger, Marsden and Rutherford, 1911)

- A narrow beam of **α-particles** (positive, = helium nuclei, 2 protons + 2 neutrons) was aimed at an extremely thin gold foil.
- Thomson's model predicted the particles would pass straight or deflect slightly. Observed: **most passed undeflected; some were deflected at large angles; a few bounced back.** Deflection is called **scattering**.

Observation	Conclusion (Rutherford)
Most α-particles passed straight through	Most of the atom is **empty space**
Some deflected through large angles	A positively charged centre repels the positive α-particles
A very few rebounded	Positive charge and most of the mass are packed in a tiny, dense **nucleus**

IMPORTANT

Rutherford's planetary model: the nucleus holds all the positive charge and most of the mass; electrons revolve around it like planets around the Sun. Nucleus diameter ≈ 10^{-15} m; atom ≈ 10^{-10} m — the nucleus is about **10^5 times smaller** (if an atom were a cricket ground, the nucleus would be a black-pepper grain). A thicker foil would scatter more α-particles because there is a greater chance of hitting nuclei; negative particles would be attracted rather than repelled.

- Number of atoms to make a paper 0.1 mm thick: $10^{-4} \div 10^{-10}$ = about 10^6 (one million).
- **Limitation:** a revolving electron constantly changes direction, i.e. accelerates; by classical ideas it should lose energy, spiral inward and fall into the nucleus — atoms would collapse. But atoms are stable.
- Rutherford (Father of Nuclear Physics; Nobel 1908 in Chemistry) also discovered the **proton**: the positive charge of the nucleus comes from protons, which are far heavier than electrons with equal and opposite charge. Neutral atoms have equal protons and electrons (He: 2 + 2; Na: 11 + 11).

▸ Bohr's model (1913)

- Electrons move in fixed circular paths — **orbits/shells/stationary states/energy levels** — K, L, M, N… or n = 1, 2, 3, 4…
- In a shell the electron has a definite energy and does **not lose energy** (postulated). This explains stability.
- K-shell (n = 1) is closest to the nucleus and has least energy; energy increases outward. Electrons exist only in allowed shells, not between them.
- An electron changes shell by absorbing or releasing a fixed amount of energy equal to the difference between the levels.
- Each shell holds a limited number of electrons. Bohr received the Nobel Prize in 1922. Letters K, L, M… come from Charles Barkla's X-ray work: he named the first line K and left earlier letters for possible lower series.
- Even Bohr's model has limits: today electrons are described as **electron clouds** (quantum mechanical model) — regions where they are most likely found.

Model	Scientist	Main idea	Limitation
Indivisible atom	Dalton (1808)	Atoms are tiny and indivisible	Atoms contain subatomic particles
Plum pudding	Thomson (1897)	Electrons in a sphere of positive charge	Could not explain α-scattering
Nuclear/planetary	Rutherford (1911)	Dense positive nucleus; electrons revolve	Could not explain stability
Shell/energy-level	Bohr (1913)	Electrons in fixed shells, no energy loss	Not fully correct; replaced by quantum model

4.3 The Neutron and Mass of the Atom

- Hydrogen has 1 proton; helium has 2 protons but is about **4 times** heavier — something else adds mass without charge.
- **James Chadwick** (1932, a student of Rutherford) discovered the **neutron**: mass ≈ proton, no charge. Found in all nuclei except ordinary hydrogen. Nobel Prize in Physics 1935.
- Electron mass is negligible, so atomic mass ≈ protons + neutrons in the nucleus.

Particle	Symbol	Relative charge	Location
Electron	e^-	−1	Outside nucleus, in shells
Proton	p^+	+1	Nucleus
Neutron	n^0	0	Nucleus

- Light atoms have roughly equal protons and neutrons (C: 6 + 6; O: 8 + 8). Heavy atoms need more neutrons (Fe: 26 p, 30 n; U: 92 p, 146 n). Neutrons reduce proton–proton repulsion by increasing the distance and strengthen the **nuclear force** that binds the nucleus.
- Neutrons, being uncharged, easily penetrate nuclei, enabling artificial radioactive elements and fission of uranium — nuclear power and weapons. BARC (Mumbai) uses neutron scattering with the Dhruva reactor to study materials.
- About 118 elements are known today (69 known by 1869); some are artificial.

4.4 Symbols of Elements

- Dalton (1803) introduced pictorial symbols; **Berzelius (1813)** proposed symbols from **Latin names**; **IUPAC** now approves names and symbols.
- Rules: first letter capital, second small (Al not AL; Co not CO); often first one/two letters; some use a different later letter (Cl, Zn); some come from Latin/Greek/German names.

Element	Symbol	Origin	Element	Symbol
Sodium	Na	Natrium	Hydrogen	H
Potassium	K	Kalium	Carbon	C
Iron	Fe	Ferrum	Chlorine	Cl
Copper	Cu	Cuprum	Magnesium	Mg
Silver	Ag	Argentum	Neon	Ne

Element	Symbol	Origin	Element	Symbol
Gold	Au	Aurum	Zinc	Zn
Lead	Pb	Plumbum	Calcium	Ca
Mercury	Hg	Hydrargyros (Greek)	Argon	Ar
Tungsten	W	Wolfram (German)	Silicon	Si

4.5 Atomic Number and Mass Number

DEFINITION

Atomic number (Z) = number of protons in the nucleus = number of electrons in a neutral atom. It uniquely identifies an element and decides its chemical behaviour. **Mass number (A)** = protons + neutrons (the nucleons).

FORMULA

$A = p + n \rightarrow n = A - Z$. Notation: ${}^{A}_{Z}X$ (e.g. carbon: mass number 12, atomic number 6).

Element	Protons	Neutrons	Mass number
Hydrogen	1	0	1
Helium	2	2	4
Lithium	3	4	7

Problem	Working	Answer
Z = 26, A = 56	p = e = 26; n = 56 − 26	26 e, 26 p, 30 n
20 protons, A = 41	41 − 20	21 neutrons
18 neutrons, Z = 17	17 + 18	A = 35
^{23}X has 11 electrons	n = 23 − 11	12 neutrons
^{70}X with 31 electrons	70 − 31	39 neutrons
79 protons, A = 197 (gold)	197 − 79	118 neutrons, 79 electrons
A = 24, Z = 12 (magnesium)	p = e = 12; n = 12	Configuration 2, 8, 2

4.6 Distribution of Electrons in Shells

1. Maximum electrons in a shell = **$2n^2$**: K (n = 1) 2, L (n = 2) 8, M (n = 3) 18.
2. Maximum in the **outermost** shell is **8** (2 if the outermost shell is K).
3. Shells fill stepwise outward: K first, then L, then M… (Bohr–Bury rules).

Element (Z)	Configuration K L M	Element (Z)	Configuration K L M
H (1)	1	Na (11)	2 8 1
He (2)	2	Mg (12)	2 8 2

Element (Z)	Configuration K L M	Element (Z)	Configuration K L M
Li (3)	2 1	Al (13)	2 8 3
Be (4)	2 2	Si (14)	2 8 4
B (5)	2 3	P (15)	2 8 5
C (6)	2 4	S (16)	2 8 6
N (7)	2 5	Cl (17)	2 8 7
O (8)	2 6	Ar (18)	2 8 8
F (9)	2 7	—	—
Ne (10)	2 8	—	—

- Neutrons in common atoms: He 2, Li 4, Be 5, B 6, C 6, N 7, O 8, F 10, Ne 10, Na 12, Mg 12, Al 14, Si 14, P 16, S 16, Cl 18, Ar 22.
- Valence electrons: C 4, F 7, Si 4. The riddle atom (A = 23, 11 protons, soft metal reacting vigorously with water) is **sodium** with 12 neutrons.

4.7 Valency

- **Combining capacity** is expressed using hydrogen or chlorine (each 1). In H_2O, oxygen combines with two H → capacity 2; NH_3 → N is 3; $MgCl_2$ → Mg is 2.
- The **valence shell** is the outermost occupied shell; its electrons are **valence electrons**. A full octet (or duplet for He) → stable, largely unreactive; incomplete shell → reactive.
- **Valency** = number of electrons gained, lost or shared to reach a stable configuration. Fewer than 4 valence electrons → lose; more than 4 → gain; carbon (4) shares.

Atom	Configuration	Valence e^-	Behaviour	Valency
Na	2, 8, 1	1	Loses 1	1
Mg	2, 8, 2	2	Loses 2	2
Al	2, 8, 3	3	Loses 3	3
C	2, 4	4	Shares 4	4
N	2, 5	5	Gains 3	3
O	2, 6	6	Gains 2	2
Cl	2, 8, 7	7	Gains 1	1
Ne/Ar	2, 8 / 2, 8, 8	8	Stable	0

4.8 Isotopes, Isobars and Average Atomic Mass

DEFINITION

Isotopes: atoms of the **same element** with the **same atomic number but different mass numbers** (different neutrons). **Isobars:** atoms of **different elements** with the **same mass number** but different atomic numbers.

Isotopes of	Names / notation	Protons	Neutrons
Hydrogen	Protium ^{1}H (~99.98 %)	1	0
	Deuterium ^{2}H (~0.015 %)	1	1
	Tritium ^{3}H (trace)	1	2
Carbon	C-12 (most abundant), C-13, C-14	6	6, 7, 8

- Isotopes have the same number of electrons and configuration, so **identical chemical properties**, but different physical properties (density, melting and boiling points).
- Uses: uranium-235 as nuclear reactor fuel; cobalt-60 for cancer radiotherapy; iodine-131 for goitre and thyroid cancer; carbon-14 to date fossils and artefacts.
- Isobars example: calcium (20), potassium (19) and argon (18) all with mass number 40. In question 3: Y (17p, 18n) and Z (17p, 20n) are isotopes of chlorine (35 and 37); Z (A = 37) and X (18p, 19n, A = 37) are isobars.

▸ Average atomic mass

- Atoms are weighed in **unified atomic mass units (u)** (earlier amu), because kg is too large.
- A simple average ignores abundance. The accurate value is the **weighted average**: Σ (isotope mass × % abundance ÷ 100).
- Chlorine: ^{35}Cl (75 %) and ^{37}Cl (25 %) → 35 × 0.75 + 37 × 0.25 = 26.25 + 9.25 = **35.5 u**. No single atom has mass 35.5 u; in 1 million atoms, 7.5 lakh are Cl-35 and 2.5 lakh Cl-37.
- Bromine: 79 × 0.497 + 81 × 0.503 ≈ **80 u**.
- Two atoms with 11 protons but 12 and 13 neutrons: same atomic number (11), different mass numbers (23, 24), same element (isotopes).
- **Homi J. Bhabha** founded TIFR and BARC and is called the father of the Indian nuclear programme. Electron microscopes (STM, TEM) let us image surfaces and arrangements of atoms.

Likely question	Model answer
Describe Rutherford's gold foil experiment and its conclusions.	α-particles on thin gold foil: most passed straight, some deflected, few bounced back, showing mostly empty space and a small, dense, positive nucleus.
List the limitations of Thomson's and Rutherford's models.	Thomson's could not explain large-angle scattering; Rutherford's could not explain why accelerating electrons do not spiral into the nucleus.
State the postulates of Bohr's model.	Electrons revolve in fixed shells (stationary states) with definite energies, do not radiate energy there, and change shell only by absorbing/emitting an energy equal to the difference.
Who discovered the electron, proton and neutron?	Thomson (1897), Rutherford, Chadwick (1932).
Define atomic number and mass number. How are they related to protons and neutrons?	Z = number of protons; A = protons + neutrons.
State the $2n^2$ rule and the maximum electrons in K, L, M shells.	Maximum in nth shell = $2n^2$; 2, 8, 18; outermost shell not more than 8.

Likely question	Model answer
What is valency? Find valency of Na, O, C.	Electrons lost, gained or shared to get a stable configuration; Na 1, O 2, C 4.
Define isotopes and give two uses.	Same Z, different A; cobalt-60 in cancer therapy, carbon-14 in dating.
Why do isotopes of an element have similar chemical properties?	They have the same electron number and configuration.
How do isobars differ from isotopes?	Isobars: different elements, same mass number (e.g. Ar-40, K-40, Ca-40); isotopes: same element, different mass number.
Calculate the average atomic mass of chlorine.	35 × 3/4 + 37 × 1/4 = 35.5 u.
Why are atoms electrically neutral?	Number of protons (+) equals number of electrons (−).

★ KEY TAKEAWAYS

- Ancient Indian (parmanu) and Greek (atomos) ideas preceded Dalton's experiment-based atom (1808).
- Thomson found the electron (plum pudding); Rutherford found the nucleus and proton (planetary model); Bohr proposed fixed energy shells; Chadwick found the neutron.
- Atomic number = protons; mass number = protons + neutrons; shells hold $2n^2$ electrons, and the outer shell at most 8.
- Valency depends on valence electrons: lose if fewer than 4, gain if more than 4, share for 4.
- Isotopes: same Z, different A, same chemistry; isobars: same A, different Z; average atomic mass is the weighted mean.

CHAPTER 05 · PHYSICS

Describing Motion Around Us

*To study the huge variety of motion in nature, physics begins with idealised forms: **linear** (straight line), **circular** and **oscillatory**. This chapter describes straight-line motion precisely with **position, displacement, average speed, average velocity and average acceleration**, shows how to read and draw **position–time and velocity–time graphs**, derives the **three kinematic equations**, and ends with **uniform circular motion**.*

Big Questions

- How much distance should we keep from the truck ahead to avoid a collision if it brakes suddenly?
- Does this distance depend on the speed at which we are moving?

5.1 Describing Position and Motion

- Position needs a fixed **reference point (origin O)**, the **distance** from it and the **direction**. In a straight line, direction is shown by signs: right of O is **positive (+)**, left is **negative (−)**.
- An object is **in motion** if its position relative to the reference point changes with time; it is **at rest** if it does not. (Hence an object on Earth is at rest relative to the ground but moving relative to the Sun.)
- An **instant** is a single clock reading; a **time interval** is the duration between two instants — not the same thing.
- We choose the origin and positive direction for convenience, but must not change them while solving a problem (e.g. origin at ground in a thrown-ball problem; origin at the release point and downward positive for a dropped object).

5.2 Distance and Displacement

	Distance travelled	Displacement
Meaning	Total length of path travelled	Net change in position between two instants (straight-line gap from start to end)
Quantity type	Scalar — magnitude only	Vector — magnitude and direction
Can be zero?	Only if no motion at all	Yes, even after a long trip (return to start)
Relation	Always ≥ magnitude of displacement; equal when motion is in one direction without turning back	
SI unit	metre (m)	metre (m)

- **Athlete example:** O → A (100 m) → back to B (40 m mark). Total distance = 100 + 60 = **160 m**; displacement = **40 m** in the positive direction.
- Ball thrown up and returning to O: at the highest point (B = 60 cm) distance = displacement = 60 cm; when back at O, distance 120 cm, displacement 0. So the magnitude of displacement is **less than or equal to** the distance, never greater.

- Fuel used depends on **distance travelled**, not displacement. A ball rolling down a curved track is not in straight-line motion; at points where it has not turned back, distance and magnitude of displacement are equal along the path only if the path is straight.

REMEMBER

Scalars (distance, speed, time) need only a number with unit; **vectors** (displacement, velocity, acceleration) need magnitude and direction.

5.3 Average Speed and Average Velocity

FORMULA

Average speed = total distance travelled ÷ time interval. Average velocity v_{av} = displacement ÷ time interval = s ÷ t. SI unit m s^{-1} (also km h^{-1}; 1 km h^{-1} = 5/18 m s^{-1}).

	Average speed	Average velocity
Based on	Distance	Displacement
Direction	None (scalar)	Same as displacement; sign + or − in a line (vector)
Can be zero?	No, if the body has moved	Yes (if it returns to start)

- Equal distances in equal time intervals (for every choice of interval) → **uniform motion** (constant speed). Unequal distances in equal intervals → **non-uniform motion** (speed increasing, decreasing or both).
- Magnitude of average velocity = average speed **only when motion is in one direction** without turning back.
- Velocity is the **average rate of change of position** with time. 'Velocity' in this chapter means velocity at an instant (**instantaneous velocity**, approached as the time interval becomes infinitesimally small). A speedometer shows its magnitude; the tyres' direction gives its direction.

Example	Working	Answer
Two postmen 210 yojanas apart walking towards each other at 9 and 5 yojanas/day (Ganitakaumudi, 14th c.)	Combined 14 yojanas/day; 210 ÷ 14	15 days (135 and 75 yojanas)
Sarang swims to the far end and back in 50 s in a 25 m pool	Distance 50 m; displacement 0	Speed 1 m s^{-1}; velocity 0
Drive 200 km north in 3 h, then 200 km south in 2 h	Distance 400 km, time 5 h; displacement 0	Speed 80 km h^{-1}; velocity 0

IMPORTANT

Magnitude of average velocity is **zero but average speed non-zero** whenever the object returns to its starting point (a round trip, a full lap).

5.4 Average Acceleration

FORMULA

a = (final velocity − initial velocity) ÷ time interval = (v − u) ÷ (t_2 − t_1). SI unit **m s^{-2}**.

- Acceleration is a vector. In a straight line: if the **magnitude of velocity increases**, acceleration is in the direction of velocity (+); if it **decreases** (retardation/deceleration), acceleration is **opposite** to velocity (−).
- Acceleration arises from a change in the **magnitude** of velocity, its **direction**, or both.
- Acceleration depends on **how quickly velocity changes**, not on how fast the object moves: a bus at a steady 100 km h^{-1} has zero acceleration.
- If velocity changes by equal amounts in equal intervals of time, acceleration is **constant**. All treatment here assumes constant acceleration. Acceleration at an instant is called instantaneous acceleration.

Example	Working	Answer
Bus speeds up from 36 km h^{-1} (10 m s^{-1}) to 54 km h^{-1} (15 m s^{-1}) in 10 s	(15 − 10) ÷ 10	+0.5 m s^{-2} (along velocity)
Same bus brakes from 15 m s^{-1} to rest in 5 s	(0 − 15) ÷ 5	−3 m s^{-2} (opposite to velocity)
Dropped object: velocities 0, 9.8, 19.6, 29.4, 39.2 m s^{-1} at 0, 1, 2, 3, 4 s	Each second change = 9.8 m s^{-1}	Constant **g = 9.8 m s^{-2}**, downward

REMEMBER

To convert km h^{-1} to m s^{-1} multiply by 1000/3600 = 5/18 (36 km h^{-1} = 10 m s^{-1}).

5.5 Graphical Representation of Motion

- Graphs show how position, velocity or acceleration depends on time. A graph is **not a route map**; it shows how the quantity changes with time.
- For motion in a straight line in one direction, distance = magnitude of displacement and speed = magnitude of velocity; if position is zero at t = 0, the position–time graph equals the distance–time graph.
- **Plotting:** draw perpendicular x- and y-axes meeting at the origin; time on the x-axis, the other quantity on the y-axis; choose a convenient scale (e.g. 5 divisions = 1 s; 5 divisions = 20 m); mark each pair of values; join the points (straight line or smooth curve). Example data: 0, 20, 40, 60, 80, 100, 120 m at 0–6 s gives a straight line; 0, 1, 4, 9, 16, 25, 36 m at 0, 2,…, 12 s gives a curve.

▸ Position–time graph

Shape of graph	Meaning
Straight line sloping up	Constant velocity (equal displacements in equal times)
Curve (steepening)	Changing velocity — accelerated motion
Line parallel to time axis	Object at **rest** (position unchanged)
Steeper line (greater slope)	Greater velocity (B steeper than A → $v_B > v_A$)

FORMULA

Slope of a position–time graph = change in position ÷ change in time = **velocity**. Example: from (2 s, 40 m) to (4 s, 80 m): (80 − 40) ÷ (4 − 2) = **20 m s^{-1}**. Parallel lines on a position–time graph mean equal velocities.

▸ Velocity–time graph

Shape of graph	Motion	Acceleration
Horizontal line	Constant velocity	Zero
Straight line rising	Velocity increases uniformly	Constant, along velocity (+)
Straight line falling	Velocity decreases uniformly	Constant, opposite to velocity (−)

- **Slope** of a velocity–time graph = (v − u) ÷ (t_2 − t_1) = **acceleration**. Rising car: (10 − 5) ÷ (20 − 10) = **+0.5 m s^{-2}**; falling car: **−0.5 m s^{-2}**.
- **Area** between the velocity–time graph and the time axis = **displacement**. Constant velocity: 20 m s^{-1} × 6 s = **120 m** (rectangle). Uniform acceleration: area = rectangle + triangle, e.g. between 10 s and 20 s: (5 × 10) + ½ × 10 × 5 = 50 + 25 = **75 m**.
- Total displacement over several phases = sum of areas of each phase (rectangle for constant velocity, triangle/trapezium for changing velocity).

From the graph	Quantity	How to find
Position–time	Velocity	Slope
Velocity–time	Acceleration	Slope
Velocity–time	Displacement	Area under line

5.6 Kinematic Equations (Constant Acceleration, Straight Line)

FORMULA

(1) v = u + at (2) s = ut + ½ at^2 (3) $v^2 = u^2 + 2as$
u = initial velocity, v = final velocity, a = constant acceleration, t = time, s = displacement.

- **Derivation of (1):** from a = (v − u) ÷ t.
- **Derivation of (2):** displacement = area under the v–t graph = rectangle (u × t) + triangle (½ × t × (v − u)); put v − u = at → s = ut + ½ at^2.
- **Derivation of (3):** from (1), t = (v − u)/a; substitute in (2) and simplify → $v^2 = u^2 + 2as$.
- Two further forms: s = vt − ½ at^2 and s = ½ (u + v) t (the area of a trapezium).
- Valid **only for constant acceleration**. Signs of u, v, a, s show direction; for motion in one direction, distance = |s| and speed = |v|.
- A body starting from rest has u = 0; a body that stops has v = 0; a retarding body has negative a.

Problem	Working	Answer
Brakes give a = −4 m s^{-2}; stopping distance at 54 km h^{-1} (15 m s^{-1}) and 108 km h^{-1} (30 m s^{-1})	0 = u^2 − 8s → s = u^2/8	28.1 m and 112.5 m (doubling speed quadruples distance)
Car from rest to 24 m s^{-1} in 6 s	a = 24/6; s = ½ × 4 × 36	a = 4 m s^{-2}; s = 72 m
Bike u = 28 m s^{-1} stops in 98 m	0 = 784 + 2a(98) → a; t = 28/|a|	a = −4 m s^{-2}; t = 7 s
Truck slows from 54 to 36 km h^{-1} in 36 s	u = 15, v = 10; s = ½ (15 + 10) × 36	450 m

Problem	Working	Answer
Car: 0→20 m s^{-1} in 5 s; 20 m s^{-1} for 10 s; stops in 6 s	50 m + 200 m + 60 m (areas)	310 m
Bus at 10 m s^{-1}; obstacle 30 m ahead; 0.5 s reaction; a = −2.5 m s^{-2}	Reaction 5 m + braking 100/5 = 20 m = 25 m	Stops 5 m short — safe
Car 6 m s^{-1} for 120 s then a = 1 m s^{-2} for 6 s	720 m + (6 × 6 + ½ × 36)	774 m

IMPORTANT

Safe following distance: stopping distance depends on speed when brakes are applied, road surface (wet/dry), braking capacity (retardation), tyre condition and the driver's **reaction time**. It rises with the square of speed, so keep a larger gap at higher speed. V2V communication technology warns drivers of collisions.

5.7 Motion in a Plane and Uniform Circular Motion

- Motion in a plane (two dimensions): overtaking vehicle, a kicked ball's path, a satellite in circular orbit. Motion in space (three dimensions): a car climbing a mountain road, a bird flying.
- **Circular motion:** motion along a circle. For a child on a merry-go-round moving from A to C: distance is the arc ABC, displacement the straight chord AC — unequal.
- In **one revolution** (period T, radius R): distance = circumference = **2πR**; displacement = **0**. So average speed = **2πR ÷ T** while average velocity over T is **0**.
- **Uniform circular motion:** motion in a circle at **constant speed**. Speed is constant but the **direction of velocity changes continuously** (imagine a runner on a rectangular, then hexagonal, then circular track: more sides → turns more frequent → circle).
- Marble-in-ring activity: when the ring is lifted, the marble moves off in a **straight line** along the **tangent** at the point of release — the velocity at any point is along the tangent to the circle in the direction of motion.
- Since velocity changes direction, uniform circular motion is **accelerated motion** even at constant speed. Everyday talk limits 'acceleration' to speed changes, but a change of direction alone also means acceleration.
- Uniform circular motion is an **idealised model** (real motion rarely has perfectly constant speed and path) but is the base for planets revolving round the Sun and vehicles taking circular turns.
- Minute hand tip (7 cm) from 6:00 to 7:30 pm: 1.5 revolutions → distance = 1.5 × 2π × 7 ≈ **66 cm**; displacement = diameter = **14 cm**; speed ≈ 66/5400 ≈ 0.012 cm s^{-1}; velocity ≈ 14/5400 ≈ 0.0026 cm s^{-1}.

Likely question	Model answer
Distinguish between distance and displacement.	Distance: total path length, scalar. Displacement: shortest straight-line change of position from start to end, vector; can be zero while distance is not.
When are average speed and magnitude of average velocity equal?	When the motion is in a straight line without turning back (one direction).
Give the SI units and formulae of average velocity and acceleration.	v = s/t in m s^{-1}; a = (v − u)/t in m s^{-2}.
Can an object with constant speed be accelerating? Example?	Yes. A body in uniform circular motion has constant speed but changing direction of velocity.

Likely question	Model answer
What does the slope of position–time and velocity–time graphs give?	Velocity and acceleration respectively.
What does the area under a velocity–time graph give?	Displacement in that time interval.
What does a straight line parallel to the time axis mean in a (i) position–time (ii) velocity–time graph?	(i) Object at rest; (ii) constant velocity, zero acceleration.
Write the three kinematic equations. When are they valid?	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$; only for constant acceleration in a straight line.
Why does stopping distance increase greatly with speed?	From $v^2 = u^2 + 2as$, $s = u^2/2\|a\|$, so doubling u gives four times s (28.1 m → 112.5 m).
Define uniform circular motion. What is the direction of velocity?	Motion in a circle with constant speed; velocity is along the tangent to the circle at that point.
A round trip gives zero average velocity. Why?	Displacement is zero as the object returns to the starting point.
Derive $s = ut + \frac{1}{2}at^2$ from a v–t graph.	Area of rectangle (ut) + triangle ($\frac{1}{2}$ t (v − u)), and $v - u = at$.

★ KEY TAKEAWAYS

- Distance is scalar (path length); displacement is a vector (net change in position) and its magnitude never exceeds distance.
- Average velocity = displacement ÷ time; average acceleration = change in velocity ÷ time (m s^{-2}); acceleration reflects how quickly velocity changes, not how fast the body moves.
- Position–time slope = velocity; velocity–time slope = acceleration; area under velocity–time graph = displacement.
- For constant acceleration: $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$.
- Stopping distance grows with the square of speed and with reaction time — keep a safe gap.
- In uniform circular motion speed is constant, velocity is tangential and changing, so it is accelerated; one revolution gives distance $2\pi R$ and displacement 0.

CHAPTER 06 · PHYSICS

How Forces Affect Motion

*Chapter 5 described motion; this chapter asks **what causes a change in motion**. It covers the nature and measurement of **force**, **balanced and unbalanced forces**, the role of **friction**, and **Newton's three laws of motion** with their everyday applications — from catching a cricket ball and airbags to walking, rowing a canoe and launching a rocket.*

Big Questions

- Why does a canoe move forward when the canoeist pushes water backwards, and faster when pushing harder?
- Will an empty canoe or one carrying a passenger move faster for the same paddle force?

6.1 The Concept of Force

- A **force** can make an object move from rest, change its speed or direction, or change its shape (kicking a ball, a bat striking a ball, squeezing a lemon).
- Force is a **vector**: it has magnitude and direction. If either changes, its effect changes. Examples with direction: friction opposes motion; magnets attract/repel along a line; the Earth pulls objects downward (gravitational force); a liquid pushes an object upward (buoyant force).
- **SI unit: newton (N)** — written with a small 'n' for the unit name (newton), capital N for the symbol.
- Magnitude can be measured with a **spring balance** (weight = gravitational force by the Earth). Smallest force we feel is about a millinewton (10^{-3} N); specialised experiments measure down to yoctonewtons (10^{-24} N).

6.2 Balanced and Unbalanced Forces

- Usually several forces act together: pushing a box (applied force + friction opposite to motion); a ball floating on water (gravitational force down, buoyant force up).
- **Tug of war:** equal and opposite pulls → **balanced forces**, rope does not move; unequal pulls → **unbalanced forces**, rope moves along the larger force.

Situation	Net force	Direction
Two forces in the same direction	Sum of magnitudes	Same direction
Two forces in opposite directions	Difference of magnitudes	Along the larger force
Equal and opposite	Zero (balanced)	None

Example (10 N and 6 N)	Net force
Both to the right	16 N to the right
10 N right, 6 N left	4 N to the right
6 N right, 10 N left	4 N to the left

IMPORTANT

An object's motion depends only on the **net force**. A weightlifter holding a barbell steady: weight (down) and her upward push are balanced. Barbell of 30 kg: F = mg = 30 × 9.8 = **294 N** upward. In arm-wrestling, when the arm tilts, the forces are unbalanced and the player who exerts more force wins.

- Forces at an angle, or equal and opposite forces on the two ends of an extended object (which make it rotate, like a tap or handlebar), are studied in higher classes.

6.3 Friction — Often Overlooked, Always Present

- Friction acts between surfaces in contact, **opposite to the direction of motion (or attempted motion)**. A box does not move until the applied force exceeds friction so that a net force acts forward.
- Other forces on a box: **weight** (gravitational, downward) and the **normal force** (exerted by the surface, upward, perpendicular to it) — these balance each other. Air also exerts a friction (drag) that is often small enough to ignore.
- A bicycle or ball stops after you stop pushing because friction acts against motion. A force is needed to *counter friction* to keep a body moving; no force is needed to keep it moving at constant velocity if friction is absent.

▸ Activity 6.1 and 6.2 — friction depends on the surfaces

Step 1

Stack four ₹10 coins with tape. Mark points A, B and C on a surface; stretch a rubber band back to C using the stack and release.

Step 2

Repeat on wood, laminated top and polished marble/tile with the same stretch. Measure the distance travelled each time.

Step 3

Result: the smoother the surface, the **farther** the stack travels and the **more slowly** its velocity decreases — friction is smaller.

Step 4

With a spring balance pulling a wooden block, the reading when the block just moves (at constant velocity) approximates the friction. Smallest reading on the surface where the stack went farthest; largest where it went least.

- Before release the stack is stationary, so forces are balanced. After release the rubber band's force exceeds friction → net force forward → acceleration. Once contact with the band is lost, only friction acts (backward) and the stack slows and stops.

▸ Thought experiment (Galileo)

- With **zero friction** on a horizontal surface, the object would never slow down and would keep moving forever. For ages it was wrongly believed that a force is needed to maintain motion; in the 17th century **Galileo** argued through thought experiments that if all impediments are removed, a body continues to move indefinitely.

6.4 Newton's First Law of Motion

DEFINITION

First law (law of inertia): An object at rest remains at rest, and an object in motion continues to move with a constant velocity, unless a net force acts on it.

- **Inertia** (Newton's word) = the tendency of objects to resist a change in their state of rest or uniform motion.
- If net force = 0, then acceleration = 0. 'Constant velocity' means no change in magnitude **or** direction (a straight line at uniform speed).
- Example: a person pushes a moving box with force equal to friction → net force zero → box continues at constant velocity.
- Graphs for zero net force: **at rest** — position–time horizontal line, velocity–time line along the time axis; **constant velocity** — position–time straight sloping line, velocity–time horizontal line above the axis.
- A real object seldom has *no* force on it; the law is achieved when forces cancel (e.g. weight balanced by normal force, applied force balanced by friction).
- Questions: an object moving with constant velocity has **zero net force**; with no net force it can stay at rest or move uniformly, but cannot have constant acceleration.

6.5 Newton's Second Law of Motion

- A force produces acceleration. Hypotheses tested: (1) for the same object, larger force → larger acceleration; (2) for the same force, larger mass → smaller acceleration.
- **Activities 6.3 and 6.4 (cart–pulley demonstration):** a falling cup of known mass pulls a cart with constant force through a thread over a pulley. Starting from rest, $s = \frac{1}{2} a T^2$. With the same distance, $a_1 T_1^2 = a_2 T_2^2$, so shorter time means larger acceleration. Doubling the pulling weight raises acceleration (a little less than double because of wheel friction); doubling the cart mass lowers acceleration.

DEFINITION

Second law: When a net force acts on an object, it accelerates in the direction of the net force; the acceleration is **directly proportional to the net force** and **inversely proportional to the mass**.

FORMULA

$a = F \div m$ or **$F = ma$** (F in newton, m in kg, a in m s^{-2}). **1 N = 1 kg m s^{-2}** — the force that gives 1 kg an acceleration of 1 m s^{-2}. Weight = **$F = mg$**, g = 9.8 m s^{-2} (≈ 10 for quick estimates), independent of the mass of the object.

- A 100 g mass resting on your palm presses with about **1 N**.
- More general form: force = rate of change of **momentum** (momentum = mass × velocity), valid even when mass changes.

▸ Applications — stretching the time reduces the force

- **Cricket catch:** the fielder draws the hands backward while catching, increasing the time to stop the ball, so acceleration and force are smaller and injury is avoided.
- **Airbags (with seat belts):** inflate into a soft cushion; the passenger's head and chest take longer to stop, so the force is reduced.
- **High-jump landing mat/sand bed, bubble wrap or hay for glassware:** lengthen the stopping time and cut the force.
- **Cracking a coconut:** it stops in a very short time on a hard surface, so the ground exerts a very large force and the shell breaks.

Problem	Working	Answer
Block 25 kg, max friction 50 N; push 50 N for 2 s	Net force 0 → balanced	Remains at rest
Same block, push 55 N for 2 s	F = 5 N; a = 5/25 = 0.2; s = ½ × 0.2 × 4	a = 0.2 m s^{-2}; s = 0.4 m
Sports car 1500 kg (v–t graph): 0 → 10 m s^{-1} in 5 s east	a = 2; F = 1500 × 2	3000 N east
Same car at constant 10 m s^{-1} for 5 s	a = 0	0 N
Same car slows from 10 m s^{-1} to 0 in 5 s	a = −2; F = 1500 × (−2)	−3000 N (towards west)
Toy car 100 g moving at constant 0.5 m s^{-1}	Constant velocity	Net force = 0
Bullet 50 g at 100 m s^{-1} stops in 50 cm	$0 = 100^2 + 2a(0.5) \rightarrow a = -10^4$ m s^{-2}; F = 0.05×10^4	500 N
Football 0.4 kg kicked to 108 km h^{-1} (30 m s^{-1}) by 800 N	a = 800/0.4 = 2000; t = 30/2000	0.015 s
2 kg object at 10 m s^{-1}, friction 7 N + 3 N opposing	Net 10 N; a = −5; s = 100/(2 × 5)	10 m
Harrow m_1 gives a_1; trolley m_2 gives a_2 under the same F	$m_1 = F/a_1$, $m_2 = F/a_2$; $a = F/(m_1 + m_2)$	$a = a_1a_2 \div (a_1 + a_2)$
Snake boat: 100 oarsmen × 200 N, 5 rowing wrongly	(95 − 5) × 200	18,000 N forward

REMEMBER

For two children on identical swings, the heavier child needs a larger force for the same initial acceleration (F = ma). Tractor/trolley and similar questions: always add the masses when objects move together.

6.6 Newton's Third Law of Motion

DEFINITION

Third law: Whenever one object exerts a force on a second object, the second object simultaneously exerts an **equal and opposite force** on the first. (Action and reaction.)

- Forces always occur in **pairs** and act on **two different objects**, so they **do not cancel**. (Two equal and opposite forces on the *same* object do balance.)
- Activity 6.5: sitting on a wheeled chair and pushing a heavy table forward, you move backward; pulling the table, you move toward it.
- Activity 6.6: two spring balances hooked together and pulled show **identical readings** — the forces are equal in magnitude.
- The law holds for **all** forces, contact or non-contact: two magnets, two charged balloons, the Earth and a fruit.

Example	Action–reaction pair
Walking/running	Foot pushes ground backward; ground (through **friction**) pushes you forward. Without friction the foot would slip. Grooves on soles and treads on tyres increase friction; wet floors, ice and water- or snow-covered roads are risky.
Climbing a coconut tree	Legs push the trunk down; friction pushes the person up. Smooth trunks with less friction are harder to climb.
Canoe	Paddle pushes water backward; water pushes paddle (and canoe) forward. A harder push gives a larger forward force and more speed (drag, currents, mass and rowing style also matter).
Balloon–straw experiment / rocket	Expelled air or gas is pushed out; the gas pushes the balloon or rocket in the opposite direction. Lift-off needs this upward force to exceed the rocket's weight.
Rocket in space	Firing the engine in the direction of motion slows it — used by Chandrayaan-3's Vikram lander for a soft landing near the Moon's south pole.
Firefighter with hose	Water pushed forward exerts an equal backward force on the hose — hence the struggle.
Spacecraft with negligible gravity	Changes its velocity by expelling gas (thrusters).
Bullet and gun	Gun recoils with force equal to that on the bullet

Problem	Working	Result
Fruit and Earth pull each other equally; why does only the fruit move?	a = F/m; Earth's mass is enormous	Earth's acceleration is negligible
Bullet 0.1 kg, gun 5 kg, force 2 N	Gun: 2/5; bullet: 2/0.1	0.4 m s^{-2} and 20 m s^{-2}
Bar magnet and compass needle	Equal forces; needle is much lighter	Needle accelerates visibly, magnet does not
Loaded cart hits identical empty cart	Third law	Equal forces on each

IMPORTANT

Equal forces ≠ equal accelerations: the accelerations differ when the masses differ. A sailor jumping forward from a boat pushes the boat backward (third law), so the boat moves backward.

6.7 Forces on a System of Objects

- Two boxes (m_1, m_2) joined by a string on a frictionless surface, pulled by force F on box 1: the string's **tension T** acts on box 1 to the left and on box 2 to the right (equal and opposite — third law).
- Treat both boxes and the string as one **system**: tension is an **internal force** and ignored; only **external forces** matter. Acceleration of the system: **$a = F \div (m_1 + m_2)$**.
- External vertical forces — weights $(m_1 + m_2)g$ and normal forces — balance each other.
- Analysing box by box gives the same result. Science is often simpler when we study the whole instead of the parts (e.g. the overall motion of a walking person).

Law	Statement	Key idea	Example
First	No net force → no change in velocity	Inertia	Box pushed with force equal to friction moves uniformly
Second	F = ma	Force causes acceleration	Cricket catch, airbags
Third	Equal and opposite forces on different bodies	Forces in pairs	Rocket, rowing, walking

REMEMBER

Newton's laws work on scales from everyday objects to planets and stars; they need modification only near very massive objects, at speeds close to light, or at atomic scales.

Likely question	Model answer
State Newton's first law. What is inertia?	An object stays at rest or in uniform motion in a straight line unless a net force acts; inertia is the tendency to resist a change in state of rest or uniform motion.
State Newton's second law and derive the unit of force.	Acceleration is proportional to net force and inversely proportional to mass, F = ma; 1 N = 1 kg × 1 m s^{-2}.
State Newton's third law with two examples.	Every force has an equal and opposite force on the other body; walking, rocket, rowing.
Why do action and reaction not cancel?	They act on different objects.
Why does a fielder pull his hands back while catching a fast ball?	Longer stopping time → smaller acceleration → smaller force and less injury.
Why are airbags provided in vehicles?	They increase the time over which the passenger stops, reducing force.
How can friction be helpful? Give two uses.	It enables walking, running and tyre grip; grooves on soles and tyre treads increase it.
Distinguish balanced and unbalanced forces.	Balanced: equal, opposite, net force 0, no change in motion; unbalanced: net force ≠ 0, causing acceleration.
Calculate the weight of a 30 kg barbell.	mg = 30 × 9.8 = 294 N.
Explain how a rocket lifts off.	It expels gas downward; the gas pushes the rocket upward with an equal force, greater than its weight.
What did the coin-stack and spring-balance activities show?	Friction depends on the nature of surfaces; smoother surface, smaller friction, longer travel.
Find the acceleration of two boxes of 4 kg and 6 kg pulled with 20 N on a smooth table.	a = 20 ÷ (4 + 6) = 2 m s^{-2}.

★ KEY TAKEAWAYS

- Force is a vector (SI unit newton); only the net force decides motion — balanced forces leave motion unchanged.
- Friction opposes motion and depends on the surfaces; it is also what lets us walk and vehicles grip.
- First law: no net force → constant velocity (inertia). Second law: F = ma; weight = mg. Third law: forces come in equal, opposite pairs on different objects.
- Increasing the time of stopping reduces force (catching, airbags, mats).
- A connected system accelerates as a = F ÷ total mass; internal forces such as tension cancel out.

CHAPTER 07 · PHYSICS

Work, Energy and Simple Machines

When forces change with time or act in complicated ways, Newton's laws become awkward. ***Work, energy and power*** *give a simpler route. This chapter defines* ***work*** *scientifically, introduces the* ***work–energy theorem***, ***kinetic and potential energy***, ***conservation of mechanical energy***, ***power***, *and the three* ***simple machines*** *— pulley, inclined plane and lever — that make tasks easier without reducing the total work.*

Big Questions

- What is the speed of a child at the bottom of a slide?
- Do two children of different masses reach the bottom of the same slide with the same velocity?
- Which slide gives the largest speed at the bottom?

7.1 Work Done by a Constant Force

- Lifting a 5 kg bag by 1 m takes a force **mg** upward over 1 m. Lifting 3 bags one after another, or together with 3 times the force, gives 3 times the work; lifting one bag to 3 m also gives 3 times the work (and a machine would use 3 times the fuel).
- So work increases proportionally with both the **force** and the **displacement**. Always state which force (agency) does the work and on which object.

FORMULA

W = F × s (s = displacement in the direction of force). SI unit **joule (J)**. **1 J = 1 N × 1 m = 1 kg m^2 s^{-2}** — the work done when a constant force of 1 N displaces an object by 1 m in the direction of the force.

- Work has no direction (it is a scalar) but can be positive or negative.
- On a **force–displacement graph**, work = **area under the graph**, even when the force varies (e.g. 10 N × 1 m = 10 J for a rectangle).

▸ When is work zero?

Situation	Reason	Example
Force F = 0	No force	Object moving with no force on it
Displacement s = 0	Nothing moves	Pushing a rigid wall; a weightlifter holding a barbell steady (muscles tire from internal energy use, but no work is done on the object)
Force perpendicular to displacement	No displacement along the force	Girl carrying a box horizontally: upward force does no work

▸ Positive and negative work

	Positive work	Negative work
Direction	Displacement **along** the force	Displacement **opposite** to the force
Effect	Energy of the object increases	Energy of the object decreases

	Positive work	Negative work
Examples	Pushing a wheelchair; lifting a dumbbell up; friction on a body that helps it move	Goalkeeper stopping a ball; lowering a dumbbell; **friction on a sliding body (the stack of coins)**

Problem	Working	Result
Goalkeeper's hand moves back 15 cm while stopping a ball with 200 N	W = 200 × (−0.15)	−30 J (negative work)
Girl lifting and lowering a dumbbell	Up: force and displacement same direction; down: opposite	Positive on the way up, negative on the way down

REMEMBER

Forces occur in pairs (third law), so both bodies do work on each other: the goalkeeper does negative work on the ball, while the ball does positive work on the goalkeeper.

7.2 The Work–Energy Theorem

- A thrown ball can knock down wickets; a raised flowerpot can damage things below. An object with the **capacity to do work** has **energy**.
- Positive work on an object increases its energy, which it can later transfer to another object (the ball hands energy to the wickets). Carrom: the striker does positive work on the white coin and the white coin does negative work on the striker, and so on down the chain.
- Energy can also transfer as **heat** (from hotter to colder bodies), by **radiation** (Sun to Earth), in electric circuits, by sound waves and by nuclear reactions.

FORMULA

Work–energy theorem: work done on an object (or system) = change in its energy. Valid also for varying forces and systems. SI unit of energy = joule. The unit is named after **James Prescott Joule**, who studied how mechanical and thermal energy are related.

7.3 Forms of Energy

Form	Meaning
Mechanical	Due to motion or position of objects
Thermal	Makes things warm or hot
Light	Lets us see
Sound	Energy of vibrations of air or other molecules
Electrical	Related to the position or motion of charges
Nuclear	Stored in the nuclei of atoms
Chemical	Stored in fuels and food as bonds between atoms

- Energy changes from one form to another: bulb (electrical → light); water heater (electrical → thermal); food (chemical → mechanical in muscles); bell (mechanical → sound).

- Further examples: truck moving uphill (chemical → kinetic and potential); unwinding watch spring (elastic potential → kinetic); photosynthesis (light → chemical); water from a dam (potential → kinetic → electrical); burning match (chemical → heat and light); fire cracker (chemical → heat, light, sound, kinetic); microphone (sound → electrical); electric bulb (electrical → light and heat); solar panel (light → electrical).

7.4 Mechanical Energy

DEFINITION

Mechanical energy = energy due to motion (kinetic) or position/configuration (potential). Mechanical energy = K + U.

▸ Kinetic energy (K)

- Energy possessed because of motion. A body at rest has zero kinetic energy. It has **no direction**; SI unit joule.
- **Derivation:** $W = F \times s = ma \times (v^2 - u^2)/2a = \frac{1}{2}m(v^2 - u^2)$. For u = 0, change in energy = kinetic energy.

FORMULA

$K = \frac{1}{2} m v^2$. Doubling the velocity makes K **four times** (K is proportional to v^2). Positive work → velocity and K increase; negative work → they decrease; zero work → K unchanged.

Problem	Working	Result
Cricket ball 0.2 kg at 154.8 km h^{-1} (43 m s^{-1})	$\frac{1}{2} \times 0.2 \times 43^2$	184.9 J
Jet of 15,000 kg stopped in 100 m by wire force 367,500 N	$\frac{1}{2} \times 15000 \times v^2 = 367500 \times 100 \rightarrow v^2 = 4900$	v = 70 m s^{-1} = 252 km h^{-1}
Masses m and 4m with same K	$\frac{1}{2}mv_A^2 = \frac{1}{2}(4m)v_B^2$	$v_A : v_B = 2 : 1$
Constant velocity at different positions	v unchanged	K does not change with position
10 kg block with K = 180 J	$\frac{1}{2} \times 10 \times v^2 = 180$	v = 6 m s^{-1}

▸ Potential energy (U)

- Energy stored by an object because of **deformation** (stretched rubber band/slingshot, bent bow, compressed spring) or by a **system of objects** because of their relative positions (two magnets, two charges, ball and Earth).
- Work done in deforming or separating is stored and released as kinetic energy when let go. In internal forces like friction, work done against them is **not** stored as potential energy.
- **Activity 7.1:** a heavy ball dropped on a sand bed from 1 m and 2 m makes a deeper dent from the greater height — greater height means greater potential energy (more work done to lift it).

FORMULA

Gravitational potential energy U = mgh (work to lift an object slowly through h with force mg; U taken as zero on the ground). Valid near the Earth's surface, where g is nearly constant (≈ 9.8, or 10 m s^{-2}). Moving horizontally at constant height does not change U.

Problem	Working	Result
Ball 200 g thrown to 10 m (g = 10)	0.2 × 10 × 10	20 J
Student 50 kg to height 72.5 m (g = 10) by lift or by stairs	50 × 10 × 72.5	36,250 J in both; U depends only on height, not on the path

▸ Conservation of mechanical energy

- For a freely falling body: at the top, K = 0, U = mgh, total = mgh. After time t: v = gt, fall = ½gt², so K = ½m g²t², U = mgh − ½m g²t². The total is again **mgh**.
- Loss in potential energy = gain in kinetic energy; **mechanical energy stays constant** if no other external force (friction, air resistance) acts.
- **Pendulum (Activity 7.2):** at extremes P and R the bob has only potential energy (mgh); at the lowest point Q only kinetic energy. It reaches almost the same height each side; it finally stops because of friction at the support and air resistance. Similarly, successive peaks of a ball-roller-coaster exhibit become lower because of friction losses.
- Total energy of an isolated system (all forms) stays constant.

IMPORTANT

Slide problem: mgh = ½mv² gives **v = √(2gh)**. The speed at the bottom depends only on the height h, not on the child's mass or the shape of the slide (friction neglected).

Example	Working	Result
Truck 10,000 kg at 20 m s^{-1} on escape ramp at 30°, sand force 50,000 N	K = ½ × 10^4 × 400 = 2×10^6 J; rise = d/2, so mgh = 50,000 d; −50,000 d = 50,000 d − 2×10^6	d = 20 m
Coconut 1.5 kg falls 10 m (g = 10)	v = √(2gh); energy 150 J; depth = 150 ÷ 3000	v ≈ 14.1 m s^{-1}; depression 0.05 m
Ball 2 kg thrown up at 20 m s^{-1}, reaches 19.4 m	K = 400 J, U = 388 J; difference	Air resistance did −12 J. Gravity does negative work on the way up, positive on the way down
Ball thrown up the same velocity on the Moon (g ≈ 1/6)	h is proportional to 1/g	6 × 8 m = 48 m

7.5 Power

FORMULA

P = W ÷ t (rate of doing work). SI unit **watt (W)**; **1 W = 1 J s^{-1}**. 1 horsepower (hp) = 746 W. Named after **James Watt**, who invented an efficient steam engine.

- Running up the stairs in 1 minute and walking up in 5 minutes involve the same work but different power.
- More work in the same time, or the same work in less time, needs more power.

Problem	Working	Power
Weightlifter lifts 75 kg by 2 m in 5 s (g = 10)	W = 75 × 10 × 2 = 1500 J; ÷ 5	300 W

Problem	Working	Power
Car 1000 kg from rest to 20 m s^{-1} in 10 s	W = ½ × 1000 × 400 = 200,000 J; ÷ 10	20,000 W
Crane lifts m to 10th floor in t; then to 20th floor in 2t	Energy doubles (2E); power = 2E ÷ 2t	Needs twice the energy, but the same power
Man (60 kg) + scooter (100 kg) vs with son (40 kg) added; same v in same time	K is proportional to mass: 160 : 200	Fuel ratio 4 : 5

REMEMBER

Raising a flag by a pulley: the work depends on the weight and height, not on speed. Doubling the speed halves the time, so **power doubles**.

7.6 Simple Machines

- The total work cannot be reduced, but a machine can make it easier by changing the **magnitude or direction** of the force.
- **Effort** = force we apply; **load** = force to be overcome. **Mechanical advantage (MA) = load ÷ effort**.
- Machines do not create energy; ignoring friction, work put in = useful work on the load (conservation of energy).

Machine	Working	MA	Notes / examples
Fixed pulley	Grooved wheel with rope; changes direction of effort (pull down to lift up)	1	Flag-hoisting
Movable pulley / pulley system	Load on movable pulley, one rope end fixed	> 1	Cranes, elevators
Inclined plane	Push a load along a slope of length L to height h: effort F′ × L = mgh	L ÷ h (> 1)	Longer, gentler slope gives smaller effort over a larger distance; ramps, winding hill roads, inclined ladders
Lever	Rigid bar rotating about a fulcrum: effort × effort arm = load × load arm	effort arm ÷ load arm	Crowbar, scissors, seesaw, spoon opening a can lid

▸ Inclined plane

- Activity 7.3 (spring balance, cart, plank): lifting vertically needs force = weight; pulling up a plank needs less force, and an even less steep (longer) plank needs still less, but the force acts over a longer distance. Work is the same (ignoring friction).
- Ramp over a 30 cm step with 40 cm width: length = 50 cm (right triangle), MA = L ÷ h = 50 ÷ 30 = **1.67**.
- Winding hill roads and inclined ladders are easier because a longer path means smaller force at each step.

▸ Lever

- Parts: **fulcrum** (pivot), **load**, **effort**; **load arm** and **effort arm** are their distances from the fulcrum.
- A small force over a large distance on the effort side gives a large force over a small distance on the load side: $\mathbf{F_1d_1 = F_2d_2}$ (work same). Balance condition (beam balance, Activity 7.5): **effort × effort arm =**

load × load arm. MA = effort arm ÷ load arm.

- Seesaw: a 15 kg child at 2 m balances a 30 kg child at 1 m (15 × 2 = 30 × 1).
- Using a spoon to open a can lid gives a long effort arm; objects to be cut in scissors are placed close to the fulcrum so that the load arm is short.

Class	Position	Examples
Class I	Fulcrum between effort and load	Scissors, crowbar, pliers, seesaw, balance scale
Class II	Load between fulcrum and effort	Lemon squeezer, wheelbarrow, bottle opener
Class III	Effort between fulcrum and load	Tongs, tweezers, broom, hammer, oar

IMPORTANT

Perpetual motion machines are impossible: real machines lose energy to friction and air resistance, so they eventually stop; energy cannot be created. Water mills (gharat/panchakki) in the Himalayas turn the potential energy of water to kinetic energy to rotate a wheel that grinds grain; dams generate electricity the same way.

- Scientists have proposed **dark energy**, a mysterious form that cannot be exchanged with other forms, to explain the accelerating expansion of the Universe.

Likely question	Model answer
Define work. State its formula and SI unit.	Work = force × displacement in the direction of force; joule (1 J = 1 N m).
When is work done zero? Give three cases.	F = 0, s = 0 (pushing a wall), or force perpendicular to displacement (carrying a box horizontally).
When is work negative? Give an example.	When displacement is opposite to force, e.g. goalkeeper stopping a ball, friction on a moving body.
State the work–energy theorem.	The work done on an object equals the change in its energy.
Derive K = ½mv².	W = F s with F = ma and s = $(v^2 - u^2)/2a$ gives W = ½m$(v^2 - u^2)$; with u = 0, K = ½mv².
Derive U = mgh.	Lifting slowly needs force mg through height h, so work and stored energy = mgh.
What happens to K if velocity is doubled?	It becomes 4 times.
State the law of conservation of mechanical energy with an example.	Without external forces K + U is constant; in free fall U decreases by the same amount as K increases.
Define power, unit, and 1 hp.	Rate of doing work, P = W/t, watt (J s^{-1}); 1 hp = 746 W.
What is mechanical advantage? What is it for a fixed pulley?	Load ÷ effort; 1 (it only changes direction).
State the principle of a lever.	Effort × effort arm = load × load arm.
Why can a machine not give us extra work?	Work done on the load cannot exceed the work put in; energy cannot be created, and friction wastes some.

★ KEY TAKEAWAYS

- Work W = F × s (joule); no work if the force is zero, there is no displacement or the force is perpendicular to displacement; it can be positive or negative.
- Work done on an object equals the change in its energy (work–energy theorem).
- Kinetic energy = ½mv²; gravitational potential energy = mgh; mechanical energy (K + U) is conserved when only gravity acts.
- Speed at the bottom of a frictionless slide, v = √(2gh), depends only on height.
- Power = work ÷ time (watt); 1 hp = 746 W.
- Simple machines (pulley, inclined plane, lever) change the magnitude or direction of effort; mechanical advantage = load ÷ effort; total work is never reduced.